Bally

Bally Star Trek Pinball
Arduino Game Version ST2026.04
by Dave's Think Tank

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Introduction to Star Trek ST2026.04

Star Trek ST2026.04 re-imagines the original Bally pinball game with new rules, goals, music, light patterns, and sound clips from the TV series and films! Just plug it in, and get ready to experience the game like you've never seen it before!

Important Notes!

This software requires an Arduino Mega 2560 and a WAV Trigger sound board. See "Installing, Compiling, and Uploading to the Arduino" below.

The ST2026, unfortunately, does not and cannot provide music from the Star Trek show or movies. Background sounds are therefore all set up to play Star Trek bridge sounds.

However, adding the music is easy. See the section below titled, "Adding Music Files to the WAV Trigger". There is also a step-by-step video on YouTube:
https://youtu.be/CJBlf6C11SE?si=8CX0rF0vCIO9eKpC

New Rules

Skill Shot

Each ball starts with a skill shot. The next letter you need to finish the word B-A-L-L-Y will be flashing, and is your target for the skill shot. Your reward is 3000 points, in addition to the usual 300 or 3000 points. But that's not all! Hit the 10-point rebound first, and your points are doubled. It's possible to hit the rebound more than once, doubling the score each time! Your points earned are displayed while a Star Trek fanfare plays.

If the next letter needed is not A or B or the current saucer value, but the next saucer value, your skill shot is to hit the rebound, advancing the letters, and then land in the saucer for 12,000 points!

If two or more rebounds would be required to reach the next letter, your skill shot is again to hit the rebound and drop in the saucer for 6,000 points. Hit the rebound twice if you can!

One Ball Challenges

The three award thresholds have been reimagined in the new Arduino version of the Star Trek pinball. Instead of automatically receiving a free ball or a free game, you now receive the opportunity to win one of these awards in a one-ball challenge, plus additional points awarded throughout the challenges.

If you complete any of the three challenges, you will be given either same player shoots again, a free game, or no award, depending on your DIP-switch selection (see the section titled "Switches 14 and 15: Playfield and Scoring Threshold Special Awards" below).

Klingon Battle Challenge

Pass the first award threshold, and you are given one free ball to battle the Klingons! Hit five Klingon ships (five standing targets) for 10,000 points each, plus 50,000 for surviving the battle to the end!

Auto-Destruct Challenge

Pass the second award threshold, and you're given one ball to save the ship! Alien intruders have activated the ship's auto-destruct sequence, with a sixty second countdown. All four drop targets must be hit (for 12,500 points each) in order to turn off the auto-destruct. Your reward for saving the ship is an additional 50,000 points! Watch your time in the credit window.

Kobayashi Maru Challenge

Pass the third award threshold, and you're given one ball to take the Kobayashi Maru test, Star Trek's "no-win scenario"! You must search four planets in the system for the Kobayashi Maru (hit four drop targets for 10,000 points each, plus 10,000 for completing all four), while under attack by the Klingons. You must disable a Klingon (any of the five standing targets for 5,000 points, maximum 10) every 15 seconds or you will be destroyed. Watch your time in the credit window, and listen for the five-second warning chime.

There are areas of the playfield where you can hide from these relentless attacks, if you can find them. After finding the Kobayashi Maru, escape through the outhole without being destroyed by the Klingons and receive an additional 50,000 points. You've beaten the no-win scenario!

New Features

Fire Photon Torpedoes! Complete B-A-L-L-Y, and Captain Kirk will fire photon torpedoes from the Enterprise to celebrate the win!

Fire All Phasers! The bonus is collected at the end of each ball, or when you return to the shooter lane through the warp speed lane. The Enterprise will then fire phasers up to three times, once for each multiple earned.

Enterprise Computer: The Enterprise computer is used to calculate the match score at the end of the game, complete with sound, voice, and flashing lights!

Personal Goal: You can set a personal goal; a score that you consider a good game. If you achieve it, the game will end with an encouraging remark. If you fail, it will be less kind.

Ball Save: A particularly bad ball is not held against you! Score less than 5000 points, or play for less than 15 seconds, and Captain Kirk will allow you to "Try again."

Ball Search: If, during a game, no playfield switch is hit for 20 seconds, the game assumes the ball is stuck and conducts a ball search. This consists of firing every solenoid on the playfield, in an attempt to dislodge the ball. The search continues until every solenoid fires, or a playfield switch is hit. After all solenoids fire, if no playfield switch is hit, the tilt control is turned off, allowing you to nudge the machine to dislodge the ball from wherever it is stuck. This tilt feature can be turned off. See "Operator Game Adjustments" below.

Game Mode: This game now includes three different game modes. Game mode is selected at the start of a new game by pressing and holding the game button, either during attract mode or before player 1 begins play. After holding the button for one second, the credit window will begin scrolling the values 1, 2, and 3. Release the game button when the game you want to play is showing. The game modes available include:

1. **Regular Game Mode:** Play the standard game, based on the original Bally version but with all the new rules and features.
2. **Kids' Mode:** Similar to the regular game mode, but with all easy settings selected to make the game more fun for kids.
3. **Challenge Mode:** Three balls, and three one-ball challenges. Great for quick competitions!

Voices, lights, and music clues will let you know that you have selected the right game. If you got the wrong game, try the long-press again. Just remember that once the first player begins their game, the long-press will no longer work; the game mode will be set and you will not be able to change it. A short-press can still be used to add more players at any time during ball 1.

Start a new game, and you will be in the game mode you previously selected. This can be changed by long-pressing the game button again.

High scores are maintained separately for all three game modes. Changing the game mode will cause the high score of the current game mode to be displayed during attract mode. The different game modes can also be assigned their own threshold levels, and personal goal level. See the section titled "Setting Threshold Levels, High Score, and Personal Goal for each Game Mode" below for more information.

Expanded Self-Tests: The self-test features of the original Bally game have been significantly expanded. See the "Self-Tests" section below.

High High Scores: If your score exceeded a million, the old game would roll your score back to zero. Now high scores over a million are maintained, and displayed in the 6-digit displays(!)

New Sounds: New sound effects and voices make the game more fun and exciting. Music from the original show or bridge sounds play in the background.

Familiar Rules from the Original Game

- Making B-A-L-L-Y scores and advances lit values.
- All drop targets down scores 5000 points.
  - 1st time lites 2X bonus.
  - 2nd time lites 3X.
  - 3rd time and each additional time scores special.
- Top rollover buttons and lit flipper feed lanes score and advance hyperspace values.
- Outlanes special – Lites when B-A-L-L-Y special is lit.
- Ball thru outlane when lit scores special.
- Ball thru Warp Speed Lane (into shooter lane) scores bonus.
- Maximum 1 extra ball per ball in play.
- Tilt penalty – ball in play.
- Three threshold scores can be set to earn an extra ball or a free game[*].
- 3 or 5 balls per game.
- Above rules can be adjusted with DIP switches and self-test settings.

Game Indicators

The current version of this Star Trek pinball machine has come a long way since the original Bally version. There are more game elements, such as the skill shot and the one-ball challenges. There are multiple game modes, each with their own rules and challenges. In order to help keep track of everything, use has been made of game indicators. These include the lights, light patterns, sounds, and numeric displays. Some of the more useful indicators are reviewed below.

Credit Display

The credit display serves a number of purposes during the game:

Purpose	Explanation
Photon Torpedoes	Points received are displayed when photon torpedoes are fired.
Phaser Bonus Score	The phaser bonus received is displayed as the phasers fire.
Auto-Destruct Timer	The sixty-second countdown timer is displayed during the Auto-Destruct challenge.
15-Second Attack Timer	The time until a Klingon attack is displayed during the Kobayashi Maru test.
Match Calculation	Credit display and Match display work together, scrolling values used in match calculation.
Choose game mode	Press and hold the game button at the start of a game to choose a different game mode. Modes 1 to 3 scroll through the credit display.

[*] Although now you must complete a one-ball challenge to receive the award.

Player Displays

The player displays are primarily used to show the score for each player, as you would expect. However, when skill shot points are awarded, the points awarded are briefly shown in the player displays. If points are awarded in the saucer, the display will remain until the ball is ejected at the end of the fanfare. If points are awarded from the B or A lane, the points will remain for a minimum of two seconds.

Flashing Lights

Flashing lights generally indicate an important target. Flashing lights are primarily used in the skill shot to indicate the skill shot target, and in the one-ball challenges to indicate the targets for the game.

Solid on or off lights, when they are associated with a switch, generally indicate the number of points awarded for hitting the switch. “On” is generally more valuable than “off”. The number of points awarded will match the values printed on the playfield, if any.

Light Patterns

Pattern	Indicates
Photon torpedoes	BALLY completed, and indicated reward is being added to your score.
Phasers	Collected bonus is being awarded. Phasers fire one, two, or three times to indicate multiplier.
Enterprise Explodes	Challenge goal was not achieved.
Winner Display	Challenge goal was achieved.
Enterprise computer	Calculating match number.

Music

If you don’t have the music, see the section below titled, “Adding Music Files to the WAV Trigger”. There is also a step-by-step video on YouTube: https://youtu.be/CJBlf6C11SE?si=8CX0rF0vCIO9eKpC

Music	Plays During
The Original Series	Music from the original series, as well as bridge sounds, are played during regular game play.
Wrath of Khan	Music from this, and other Star Trek films, is used during the one-ball challenges to create an appropriate level of tension!
Fanfare from The TV Show	Indicates a win during skill shot or one of the challenges.

Voices / Sound Effects

There are lots of voice and SFX clips from the show in the game. Most indicate you have won something, or achieved something, or have an opportunity. Here are a few you should know:

Sound	Indicates
Try Again	Ball save: rescuing you from a disastrous ball!
Sensors picking up a Klingon battle cruiser	You have reached challenge 1.
One Minute to Auto Destruct	You have reached challenge 2.
Captain, I'm getting something on the distress channel	You have reached challenge 3.
Explosions during Challenge 1	You have destroyed one of the five blinking Klingon ships.
Computer sequences during Challenge 2	You have hit one of the four blinking computer keys.
Go Sulu! During Challenge 3	You have found the Kobayashi Maru. Now get out!
Other quotes during Challenge 3	One more planet eliminated in your search for the Kobayashi Maru.
Single Chime	5-second warning in the Kobayashi Maru test.
Pleasant end-quote	e.g., "Your logic was impeccable, Captain". Indicates you have exceeded your personal goal.
Unpleasant end-quote	e.g., "He's dead, Jim". Indicates you have failed to achieve your personal goal.

Installing, Compiling, and Uploading to the Arduino*

Hardware Requirements

Purchase Arduino with WAV Trigger: pinside.com/pinball/shops/shop/1304-roygbev-pinball/13777-star-trek-arduino-upgrade-with-new-rules-and-sounds
Or Build Your Own: www.pinballrefresh.com/blog/how-to-install-on-your-machine
Purchase WAV Trigger: www.robertsonics.com/wav-trigger (if not included above)
Purchase a Computer Speaker Bar, with 3.5mm stereo jack and USB power cable

Software Requirements

Download ST2026p04: https://github.com/DavesThinkTank/Bally-Star-Trek
Download Arduino IDE: www.arduino.cc/en/software (free, but there may be a donation request)
Sound Files: https://drive.google.com/drive/folders/1jDBnFHythNCg0qE2jMhNxCWICuR75obn

This software does not come with music from the show. However, there is a step-by-step video on YouTube that shows you where to obtain this music, and how to add it to your Arduino:

https://youtu.be/CJBlf6C11SE?si=8CX0rF0vCIO9eKpC

See also the section titled "Adding Music Files to the WAV Trigger" below.

Note that there are regular updates to the Bally Star Trek software. Check for updates at the GitHub link above. The steps to update your software are the same as for installation, covered below.

Hardware Installation

Install the Arduino as per instructions included with the device (Note: The software can be compiled and uploaded to the Arduino as per instructions below before installing the Arduino in the pinball, if you prefer). Detailed installation instructions are also included at the end of this manual*.

Install WAV Trigger as per instructions included with the device. Detailed installation instructions are also included at the end of this manual*. If you purchased a WAV Trigger separately (i.e., not as part of a kit from RoyGBev), you will need to review these instructions!

* See also the sections titled *Arduino Installation Instructions* and *Additional Installation Instructions* below.

Software and Sound File Installation

The following video demonstrates how to install or upgrade the Bally Star Trek software and sound files: https://youtu.be/r4PpJglYBh4

If your Arduino came with the ST2026.04 software included, and you have no need to make changes to the software, you're done! Otherwise, perform the following instructions to upload the software to your Arduino.

Create a folder on your computer named ST2026p04. Download the software from github.com/DavesThinkTank/Bally-Star-Trek to this folder, by clicking on the green "Code" button, then clicking on "Download ZIP". Note that the folder name MUST be the same as the name of the .ino file in the download or the IDE will give you an error!

Download and install the Arduino IDE from www.arduino.cc/en/software.

The sound files can be found at:
https://drive.google.com/drive/folders/1jDBnFHythNCg0qE2jMhNxCWICuR75obn

Transfer the sound files to the micro-SD card for your WAV Trigger board (into the root directory).

This software does not come with music from the show. However, there is a step-by-step video on YouTube that shows you where to obtain this music, and how to add it to your Arduino:

https://youtu.be/CJBlf6C11SE?si=8CX0rF0vCIO9eKpC

See also the section below titled "Adding Music Files to the WAV Trigger" for information on adding Star Trek music files.

Compiling and Uploading the Software

Before compiling and uploading the software to the Arduino, be sure to review the "Operator Game Adjustments." See the section by this name below. There are several options you may need to set, such as whether or not you have the music files installed, or the time limit to be used for the Kobayashi Maru challenge.

TURN OFF your pinball machine! (if programming the Arduino after installation). Make sure "Switch" connectors on the Arduino are connected with a jumper.

Plug the Arduino into a USB port on your laptop computer with an appropriate cable. You need a data cable (not a power cable!) with a USB micro connector on one end, and a USB A or C on the other end, whichever your computer accepts. Ignore the LED lights on your pinball's circuit boards. It's normal for the Arduino to power some LEDs.

Run the Arduino IDE software. Open the file ST2026p04.ino (in the ST2026p04 folder created above).

Click on the box at the top left of the IDE labeled “Select Board”. Click on “Select other board and port”. Select “Arduino Mega or Mega 2560”, and the port your Arduino board is plugged into. If you’re not sure which port to select, unplug the Arduino from your computer. The correct port is the one that disappears!

Click on the Verify checkmark (to the left of the “Select Board” box) to make sure the software compiles properly.

Click on the Upload arrow (also to the left of the “Select Board” box) to compile the software and upload to the Arduino.

Once the program has compiled and uploaded you can unplug the cable from your computer. Unplug the old speaker from the sound board in the backbox! The game should now run on your pinball.

Before You Play

When you turn on the pinball, the first thing you should see is the version number; 2026 in the player 1 display, and 04 in the credit window. This will last for about four seconds. If you don’t see this, you haven’t got the software running yet! Also, if it’s not the latest version number, go back to GitHub and find the latest release.

RUN THROUGH ALL THE SELF-TESTS AND GAME SETTINGS BEFORE PLAYING YOUR FIRST GAME! Make sure everything is set the way you want it. See the self-test section below.

Read the manual, and the readme.md file, and watch the following YouTube videos for more assistance:

Overview of New Features: https://youtu.be/NWBjEo7yrWA
Installation Assistance: https://youtu.be/-5JYS4kdSAM

Adding Music to the Arduino: https://youtu.be/CJBlf6C11SE?si=XrKO_I5-LyZGp0f7
How to Update Your Software: https://youtu.be/r4PpJglYBh4?si=fmr99n3UeNldb_0T

Self-Tests and Adjustments with the Arduino

The Arduino self-test adjustments are similar to the regular Bally adjustments, but have been significantly expanded. You begin by pressing the **red self-test button** inside the coin door.

There are getting to be a LOT of tests, audit settings, and game settings. Having to go through all these settings every time just to get back to attract mode can be annoying, and can also cause the self-test button to wear out! You can now use the **slam switch** (on the inside of the coin door) to end self-test at any point (other than during the switch test).

The Game Button, and "Any Other Switch"

In the original Bally pinball self-test, there was only one switch available to run the tests – the **game button**. Pressing this button generally stepped through the current test, increased a value, changed a setting, etc. But whatever it did, it did just that one thing, severely limiting what the self-test and game settings could do for you.

If only there was just one more button, the tests could be expanded significantly. For comparison, most modern pinball machines have a panel of four extra buttons for these tests. In order to expand the tests, it was therefore decided to add one more switch.

Deciding which switch to use wasn't easy. A switch that is convenient for one person on one machine may be completely inaccessible for another person or another machine. For this reason, it was decided that the second switch would be **"any other switch"**. That is, any switch on the playfield, in the cabinet, or on the coin door[*] can be used to provide an extra function where needed.

The self-tests have been reviewed with the goal of making maximum use of the game button, and minimizing the need for any other switches. Through single-click, double-click, and holding of the game button, most tests and data entry screens are now covered, without the need for any additional switches. There are now only a couple of places where a second switch is required. All the original tests, and the extended tests, and all data entry can be handled with just the game button, with another switch only required in a couple of special situations. The instructions below should make clear where functionality has been expended through the use of "any other switch".

[*] other than the game button, the slam switch, or the self-test switch

Self-Tests

Test the various features of your pinball machine.

Light Test

Ball in Play: Test #01
Display #1: Light number, or 99 for all lights.
Game Button: Cycle through switched Illumination lights. Hold to cycle continuously.
In pattern display screens, changes to the next displayed pattern.
In nacelle display, changes to the next nacelle pattern. Hold to cycle.
Double-Click: Change between flashing light display and light pattern displays.
Change to nacelle light patterns, while either nacelle light (#11 or 12) is lit.
Any Other Switch: Repeat the current light display pattern.

The first test will repeatedly flash all the switched illumination lights on the playfield and in the backbox. This is similar to the regular Bally light test, except the Arduino allows you to now press the game button on the front of the coin door. When you do so, all the lights will go out except one. By continuously pressing the game button, the pinball will step through all the lights, displaying each, one at a time. Display #1 shows the corresponding light number. A table of all the lights and the sequence in which they appear is included below.

Double-clicking the game button shows you different light display patterns from the game. Press the game button to step through the displays. Press any other switch while on these screens to repeat the display. Be sure to wait for the lights and sound of one display to finish before starting another, or the lights may get out of synch with the sound. In particular, the photon torpedo display has an echo effect that lasts several seconds.

The following light displays are included:

0. Flash all lights / step through one light at a time
1. Photon torpedoes
2. Phasers
3. Enterprise explodes
4. Challenge winner display
5. Enterprise computer (match display)
6. Attract display

Double-press the game button at any time to return to the flashing light screen. Note that, if you do this while a display is in process, some of the lights may not flash.

To view the nacelle light patterns, press the game button until light 11 or 12 is lit (i.e., one of the nacelle lights). Then double-click the game button. Press the game button to cycle through all nacelle light patterns. Complete all patterns, or double-click the game button to return to the regular light tests. (Obviously, the nacelle light patterns can only be viewed if you have added nacelle lights! See the section below, “Adding Nacelle Lights”.)

Display Test

Display #1-5: All digits cycle through numbers 0-9
Game Button: Cycle through individual digits. Hold to cycle continuously.
Double-Click: Change between digits 0-9 and all 8s

Pressing the self-test button again will then take you to the display test. Again, this is similar to the Bally display test in that it cycles all digits in all five displays through the numbers from 0 to 9 repeatedly. It cycles quite a bit faster than the Bally test though, making this a much less tedious review!

The Arduino extends this test with use of the game button. When you press the game button, all displays will go blank except for the first digit on the first display, which will continue to cycle. Pressing it again moves this to the second digit. Pressing it again moves to the third, and so on, going through each digit of each display individually. After the final digit, pressing the game button will set all displays running through the numbers again. Holding the game button down will cycle quickly through each individual digit.

Double-clicking the game button also extends the usefulness of the display test. By double-clicking, the test will stop cycling through the numbers 0-9, and will instead display the value 8 in all digits. This is helpful in finding missing or dim segments, and assessing burnt segments. Pressing the game button scrolls the number 8 through all digits, and double-clicking changes back to the digits 0-9.

Solenoid Test

Ball in Play: Test #03
Credit Display: Switches firing due to solenoid activity (if any)
Display #1: Solenoid number
Display #4: Time in milliseconds between solenoid firing and switch activating
Game Button: Fire current solenoid repeatedly. Press again to continue cycling.
Double-Click: Stop solenoids from firing. Also clears switch display (credit window).

Pressing the self-test button again takes you to the solenoid test. This runs through all the solenoids, just like the regular Bally test (except in a different order). See the table below for a list of solenoids, and the order used. Note, the Coin Door Lockout and the K1 Flipper Relay alternately either turn off or on, each pass through the solenoids.

Pressing the game button at any point will cause the current solenoid to continue firing repeatedly, so you no longer have to cycle through all of the solenoids to see the one you are interested in. Press again to continue cycling. Double-click to turn firing of solenoids off and back on. This allows you to make adjustments to a solenoid while remaining in test mode!

Keep an eye on the credit window during this test. If vibration from a solenoid causes a switch to misfire, the switch number will be displayed here. The time between the solenoid firing and the switch activation is displayed in Display #4 (in milliseconds). See the section "User Programmable Changes" to see how this information can be used to fix this issue.

Flippers can be tested while solenoid 15 is activated (the K1 flipper relay). Once activated, hold either flipper button to see that flipper activate and deactivate. Any switches fired through vibration from the flippers will show up in the credit window.

Stuck Switch Test

Ball in Play:	Test #04
Credit Display:	The number of switches currently closed
Display #1-4:	The lowest four stuck switch numbers
Game Button:	Double-click to reset the drop targets

Pressing the self-test again takes you to the switch test. Switches that are stuck on will be identified by number in the displays, with up to four stuck switches identified on four displays. The number of closed switches is also displayed in the Credit display, for cases where more than four switches are closed at once. The same numbering system is used as the original game, as in the table below.

Double-clicking the game button will reset all drop targets during the switch test. This allows you to easily test and work with drop target switches, and then reset them.

Note: In order to allow testing of the slam switch and game button, the special functions assigned to these buttons during self-tests do not work during the switch test.

The flippers are enabled throughout the solenoid test. This is therefore a good place to work on flipper issues.

Detecting Switch Matrix Issues

The Stuck Switch test can also be used to locate switch matrix issues. The 40 switches of a pinball are wired together in an 8x5 grid. Diodes on each switch make sure one switch closing cannot affect any other switch, but a bad diode can cause problems. If a closed switch has a bad diode, and another switch in the same row is closed, and another in the same column is closed, then a fourth switch at the opposite intersection of the row and column will also register as closed.

Testing for switch matrix issues:

1. Fix all stuck switches. Make sure all switches are open. All four displays should be blank.
2. Test all switches individually. Make sure you know where they all are. Note that the switch matrix diagram in your schematics may be inaccurate. Note any errors.
3. Start with switch 0. Close the switch, and hold it closed.
4. Choose any other switch in the same row (refer to the switch matrix chart below). Close the switch, and hold it closed.
5. Choose any other switch in the same column. Close the switch, and hold it closed.

6. Three displays should show the three switches you are holding closed. If a fourth display indicates another switch, then switch zero has a bad diode and is causing a switch matrix error. (Note, the coin slot switches do not have diodes and should register as causing a switch matrix error.)
7. If multiple switches are assigned the same switch number, be sure to test them all by opening the one you are holding, and closing the next one.
8. Open all the switches. Proceed to test switch 1, then 2, and so on to switch 39.

Switch Bounce (Double-Hit) Test

Ball in Play: Test #05
Display #1: Most recent switch hit
Display #2: The time between hits in milliseconds
Game Button: Double-click to reset all drop targets

Pressing the self-test button again takes you to the switch bounce test. Switches on your pinball machine may develop a "bounce", where hitting them registers two or more hits. If you suspect this may be happening with a switch on your machine, this test can help you to identify the issue.

To determine whether a switch is bouncing, activate the suspected switch with a pinball. If it registers only once, the switch number will appear in the Player 1 display, and all other displays will be blank. If it registers two or more times, the time between hits will appear in the Player 2 display (measured in milliseconds). See the section "User Programmable Changes" to see how this information can be used to fix this issue.

Double-clicking the game button will reset all drop targets during the switch test. This allows you to easily test and work with drop target switches, and then reset them.

Sound Test

Ball in Play: Test #06
Display #1: Sound number
Game Button: Play same sound repeatedly. Press again to continue cycling.
Press within 1 second of display change to skip current sound.
Hold to skip many sounds quickly.

Pressing self-test again takes you to the sound test. The original Bally test simply played a single sound. The Arduino cycles through all the sounds. Pressing the game button plays the current sound repeatedly. Pressing it again will continue cycling sounds.

Display #1 will indicate the sound number to be played. If the game button is pressed within one second of the display changing, the current sound will be skipped. Holding the button will increase speed, skipping sounds (very useful for the long, empty stretches!). See the table below for a list of sounds.

Each sound will take five seconds before proceeding to the next. This prevents the sounds from running over each other. Also, many sounds are empty. In particular, the first few make no sound.

DIP Switch Test

Ball in Play: Test #07
Display #1 - 4: DIP switch values (1 = ON, 0 = OFF), first six digits of 4 DIP banks
Credits: DIP switch values for final two digits of the current DIP switch bank
Game Button: Move to next DIP switch. Hold to cycle through switches quickly.
Double-Click: Change setting of current DIP switch

Pressing self-test again takes you to the DIP switch test. This completely new test shows you the setting of all 32 DIP switches, and allows you to change them until the pinball is turned off. Turning the machine off and on again restores the DIP switches to the settings on the MPU board.

All 32 DIP switches are shown in the 26 (!) display digits as either 1 (ON) or 0 (OFF). Since the displays are only six digits, the first six switches of each bank of eight are shown in the four displays. The 7^{th} and 8^{th} digits of the current bank are shown in the Credit window.

The current switch is identified by a flashing number, and the two switches in the Credit display belong to this bank. By pressing the game button, you can scroll through switches 1 to 32. The values in the Credit window will update to show the last two switches of the current bank, as the current bank changes. Stop on a switch and you can double-click the game button to change its setting temporarily. This new setting will continue to be used until the pinball is turned off, or it is reset using this screen again.

This can be useful to detect defective DIP switches, check DIP settings without opening the backbox, or to try out different DIP switch settings.

Setting Threshold Levels, High Score, and Personal Goal for each Game Mode

Credit Display: 01, 02, 03 for threshold levels 1, 2, and 3
04 for high score
05 for personal goal
Displays #1-3: Values to be set for game modes 1, 2 and 3
Game Button: Move to next display
Double Click: Set current value to zero
Hold: Increase value by 1000s. Process speeds up as you continue to hold.
Any Other Switch: Change method to set value one digit at a time

Set Current Value One Digit at a Time:

Game Button: Increase current digit. Hold to increase continuously.
Double-click to move to next digit
Return to increase-by-1000 method by moving past the ninth digit
Any Other Switch: Move to next digit (same as double-click above)
Double Click: Return to increase-by-1000 method

Pressing the self-test button again takes you to the first threshold level. There are three threshold levels for each game mode at which you can earn extra balls or free games. Three values can be set in displays 1 through 3, representing the first threshold level in game modes 1 through 3. The value in display #1 will flash for about three seconds, indicating it is the value you are currently editing. As you move between the three values, the active value will flash for three seconds so you know which value you are editing.

Pressing the self-test button again takes you to a similar screen for the second threshold level. Next is the third threshold level, then the high score, and then "Personal Goal"; a new feature that provides a surprise ending when the set goal is achieved. Set Personal Goal to what you consider a good game for each game mode.

Press the game button to move between the values on the screen.

Holding the game button increases the active value by 1000, slowly at first, and then with increasing speed. Release it to stop, and press again to start off slowly. Double-pressing the button resets the value to zero. A threshold level of zero means no award for this level or higher.

These values can also now be increased, or decreased, one digit at a time. This is a very simple alternative to the previous method, which has been in place since the 1970s. Press any switch in the pinball, other than the game switch or the slam switch, to go to the first digit. The 10s digit will flash. Press the game button, or hold it, to increase the value. After reaching 9, it will circle around to 0. Set the digit you want, then double-click the game button again to go to the next digit. Or pressing any other switch will also move you forward to the next digit.

Once you exceed the 100,000s digit, the numbers will shift right to allow you to change values in the higher digits. You can enter values up to 999,999,990. If you try to go to the 10th digit, you will be returned to the increase-by-1000 method. Or double-click any switch to return at any time.

For anyone who has spent 10 minutes or more holding in the game button while watching these values scroll, or held the game button an instant too long and had to start over, or tried to find kludgy ways to reproduce a high score, especially the lower digits, this new method of setting score levels will be a welcome relief!

Score Levels

01: Threshold Level 1 (Klingon Battle Challenge)
02: Threshold Level 2 (Auto-Destruct Challenge)
03: Threshold Level 3 (Kobayashi Maru Challenge)
04: High Score
05: Personal Goal

Note: There are no threshold levels in game mode 3, the Challenge Mode. Any values you enter in display #3 for the three threshold levels will therefore be ignored.

Accounting Info

Credit Display: Level #06 through 12
Display #1: Value of accounting item
Game Button: Increase value by 1. Hold to increase repeatedly.
Double Click: Set value to zero

Tests 06 through 12 cover number of credits, total number of games played, total number of free games won, number of times high score beat, and number of coins collected in chutes 2, 1, and 3. Hold the game button to increase, or double-click to set these to zero.

Accounting Items

06: Credits
07: Total plays
08: Total replays
09: Number of times high score beat
10: Chute 2 coins
11: Chute 1 coins
12: Chute 3 coins

Game Settings

Tests 13 and 14 change certain game settings, as outlined below.

Credit Display:	Game Setting Number (13 through 14)
Display #1:	Value of setting
Game Button:	Increase value. Returns to minimum value after reaching maximum.
Double Click:	Set value to zero

13: Background Sound On / Off
00: Background music or sounds will be silenced
01: Background will play

14: Free Play
00: No free play. Coins must be inserted to play game.
01: Game can be started by pressing the game button, without inserting coins.

If Free Play is selected, the amount in Credits (accounting item 6 above) will determine whether or not the credit light on the apron is lit. Accounting item 6 will then be the only way to turn this light on or off. This could be important, if you have a custom apron which incorporates the credit light into its design!

15: Ball Save Settings

Credit Display:	Level #11
Display #1:	Number of ball saves per game
Display #2:	Maximum score to invoke ball save
Display #3:	Maximum play time to invoke ball save
Game Button:	Move to next displayed value
Double-Click:	Reset to zero
Hold:	Increase value

“Ball Save” is a feature that will let you play a ball over again, under certain circumstances. This screen allows you to set the number of ball saves allowed per game (maximum 5), the maximum score at which ball save will be invoked (maximum 25000), and the maximum play time at which ball save will be invoked (maximum 25 seconds).

If you have a particularly bad ball, your score will be reset to the beginning of the ball, “Same Player Shoots Again” lights will come on, and Captain Kirk will offer you to “Try Again”.

If you do not want ball save, set the number per game to zero, but leave the other values non-zero! Setting all three values to zero will cause the program to think you have not set any values yet, and it will reset them to the default values.

16: Relative Volume Settings (Background Music, Sound Effects, and Voices)

Credit Display:	Level #12
Display #1:	Volume adjustment for background sounds (0 – 100)
Display #2:	Volume adjustment for sound effects (0 – 100)
Display #3:	Volume adjustment for voices (0 – 100)
Game Button:	Move to next displayed value
Hold:	Increase or decrease value
Double-Click:	Switch between increasing or decreasing values

This test allows you to adjust the relative volumes of the background music (display 1), sound effects* (display 2), and voices (display 3). In general, you want sound effects to be quieter, voices to be louder, and background music to be somewhere in the middle. The volume levels have already been set in this way, and to leave them alone, simply set all three values to 50. Values above 50 will raise the volume for sounds of that type, and values below 50 will lower those volumes.

Please note this screen is used to set the *relative* volume for the three sound types! It therefore makes little sense to raise all the values above 50, or to lower them all below 50. To raise or lower all three, you are far better off using the volume control on your speaker!

Also keep in mind, raising the volume too high on one or more sound types could cause a problem called “clipping”, where the highs and lows of the sound are lost or distorted. This can be very evident if you’ve raised two or all three types too high, and they are all played at the same time and added together. So don’t do this.

A piece of background music will play throughout this test, and you can adjust its volume up or down to hear the effects. When adjusting the sound effect or voice setting, a sound effect or voice will play repeatedly over the background music, allowing you to hear the relative effects.

Note that, at first, pressing the game button will raise the volume, and holding it will raise the volume repeatedly. To lower the volume, double-click the game button. Now pressing or holding the game button lowers the volume. Double-click a second time to increase the value again.

* Some sound effects are set to loud, and will be unaffected by the sound effect setting. This includes primarily the pop bumper explosions. They are set to be slightly louder than the voice setting.

Self-Test Information Tables

The following tables can be used, together with the self-test feature, to investigate the functioning of your pinball. These tables will assist you in determining the game feature being indicated by the values displayed during the tests.

List of Arduino Self-Tests

Ball in Play Display	Credit Display	Test / Setting
1		Lights
2		Displays
3		Solenoids
4		Stuck Switches
5		Switch Bounce
6		Sounds
7		DIP Switches
	1	Score Threshold Level 1 (Klingon Battle Challenge)
	2	Score Threshold Level 2 (Auto-Destruct Challenge)
	3	Score Threshold Level 3 (Kobayashi Maru Challenge)
	4	High Score
	5	Personal Goal
	6	Credits
	7	Total Plays
	8	Total Replays
	9	High Score Beat
	10	Coins in Chute 2
	11	Coins in Chute 1
	12	Coins in Chute 3
	13	Background Sound Off or On (0 or 1)
	14	Free Play Off or On (0 or 1)
	15	Ball Save Options
	16	Relative Volume for Background Music, Voices, and Special Effects

Lights

No.	Light	No.	Light
0	Bonus 1	30	Right Flipper Lane
1	Bonus 2	31	Left Flipper Lane
2	Bonus 3	32	B Lane
3	Bonus 4	33	A Lane
4	Bonus 5	34	Klingon L #1
5	Bonus 6	35	Klingon L #2
6	Bonus 7	36	Klingon Y
7	Bonus 8	37	Right Out Lane
8	Bonus 9	38	Left Out Lane
9	Bonus 10	39	Center Y
10	Bonus 20	40	Same Player Shoots Again (Backbox)
11	Left Nacelle	41	Match
12	Right Nacelle	42	Same Player Shoots Again (Playfield)
13	Planet Special + Center 3X	43	Credit Indicator (Apron)
14	Planet 3X + Center 2X	44	Center 10
15	Planet 2X	45	Center 25
16	Hyperspace 2	46	Center 50
17	Hyperspace 4	47	Center Special
18	Hyperspace 6	48	Ball In Play
19	Hyperspace 8	49	High Game
20	Hyperspace 10	50	Game Over
21	Hyperspace Special	51	Tilt
22	Planetary Bypass Inside Lane	52	Center B
23	Planetary Bypass Outside Lane	53	Center A
24	Saucer B	54	Center L #1
25	Saucer A	55	Center L #2
26	Saucer L #1	56	1 Up
27	Saucer L #2	57	2 Up
28	Saucer Y	58	3 Up
29	300 or 3000	59	4 Up

Solenoids

No.	Solenoid
0	Not Used
1	Not Used
2	Not Used
3	Not Used
4	Not Used
5	Knocker
6	Outhole Kicker
7	Saucer
8	Left Thumper Bumper
9	Right Thumper Bumper
10	Center Thumper Bumper
11	Left Slingshot
12	Right Slingshot
13	Drop Target Reset
14	Coin Lockout
15	K1 Flipper Relay

Note: The solenoid numbering is completely different than used in the original Bally manual. In particular, the numbers 0 through 4 are unassigned.

Switches

No.	Switch
0	Drop Target “D” (Bottom)
1	Drop Target “C”
2	Drop Target “B”
3	Drop Target “A” (Top)
4	Shooter Lane Rollover
5	Game Button
6	Tilt
7	Out Hole
8	Coin 3 (Right)
9	Coin 1 (Left)
10	Coin 2 (Center)
11	
12	
13	
14	
15	Slam
16	
17	Lower Hyperspace Rollover
18	Right Flipper Lane
19	Left Flipper Lane
20	500 Points
21	Upper Hyperspace Rollover
22	10 Point Rebound (4)
23	300 / 3000 Points
24	Planetary Bypass Inside Lane
25	Planetary Bypass Outside Lane
26	Klingon Target “Y”
27	Klingon Target “L” #2
28	Klingon Target “L” #1
29	“A” Lane
30	“B” Lane
31	Saucer
32	
33	Right Out Lane
34	Left Out Lane
35	Right Slingshot
36	Left Slingshot
37	Center Thumper Bumper
38	Right Thumper Bumper
39	Left Thumper Bumper

Switch Matrix

0: Drop Target "D" (Bottom)	8: Coin 3 (No Diode!)	16:	24: Planetary Bypass Inside Lane	32:
1: Drop Target "C"	9: Coin 1 (No Diode!)	17: Lower Hyperspace Rollover	25: Planetary Bypass Outside Lane	33: Right Out Lane
2: Drop Target "B"	10: Coin 2 (No Diode!)	18: Right Flipper Lane	26: Klingon Target "Y"	34: Left Out Lane
3: Drop Target "A" (Top)	11:	19: Left Flipper Lane	27: Klingon Target "L" #2	35: Right Slingshot
4: Shooter Lane Rollover	12:	20: 500 Points	28: Klingon Target "L" #1	36: Left Slingshot
5: Credit/Game Button	13:	21: Upper Hyperspace Rollover	29: "A" Lane	37: Center Thumper Bumper
6: Tilt	14:	22: 10 Point Rebound (4)	30: "B" Lane	38: Right Thumper Bumper
7: Outhole	15: Slam (3)	23: 300 / 3000 Points	31: Saucer	39: Left Thumper Bumper

Notes:

1. The coin 1, 2, and 3 switches do not have diodes. This means, if used during a game or stuck, they could cause a switch matrix issue. They are also handy for testing and understanding switch matrix issues for this reason.
2. The columns are labeled ST 0 through 4 (ST for strobe), and are wired to the MPU board through connector pins A4J2-1 through 5.
3. The rows are labeled I 0 through 7 (I for input), and are wired to the MPU board through connector pins A4J2-8 through 15.

Sounds

Arduino	Sound	Arduino	Sound
0		30	Kirk: Beam me up Scotty!
1		31	Scotty: Hello Computer
2		32	McCoy: You're out of your Vulcan mind!
3		33	Scotty: The warp drive is a hopeless pile of junk
4		34	Spock: Your logic was impeccable Captain
5		35	Spock: Vulcans never bluff
6		36	Kirk: Standard orbit Mr. Chekhov
7		37	
8	Beep 3- 10 Chime	38	
9	Beep 44 - 100 Chime	39	
10	Computer beep - 1000 Chime	40	
11		41	Small explosion 1
12		42	Small Explosion 2
13	Command Sequence 1	43	Small explosion 3
14	Command Sequence 2		
15	Command Sequence 3	…	
16	Command Sequence 4		
17		51	Kirk: Scotty, Energize
18		52	Kirk: James Kirk, commanding the Starship Enterprise
19	Power Up	53	Kirk: Fully Operable Scotty?
20	{Computer: Working...	54	Scotty: Prepare to take us out of orbit
21	Fire all phasers: bonus x 1	55	Uhura: It's a big galaxy, Mr. Scott
22	Fire all phasers: bonus x 2	56	Transporter (Louder)
23	Fire all phasers: bonus x 3	57	
24	One Minute to Auto Destruct	58	
25	Bridge to Captain	59	
26	Fascinating	60	
27	Fool Me Once...	61	Kirk: Ahead warp factor 1
28	If My Granny had wheels...	62	Kirk: Full ahead warp factor 1
29	I Can't Change the Laws of Physics	63	Kirk: Full ahead Mr. Sulu

Sounds Pg 2

Arduino	Sound	Arduino	Sound
64	Mr. Spock full ahead warp factor 1	110	
65	Warp 1 Mr. Sulu	111	
66	Kirk: Arm photon torpedoes	112	
67		113	Khan! (Louder)
68		115	Kobayashi Maru Intro
69		116	We need power!
70		117	We're running out of time…
71	Photon torpedoes fire, 3 quick	118	We're not going to make it
72		119	Go Sulu!
73		…	
74		122	Uhura: Calling Klingon vessel
75		123	Phasers on stun
76	We may be here for a very long time	124	Prepare to transport
77		125	Try again
78		127	
79		128	
80		129	
81	He's intelligent but not experienced	130	Sulu: Intruder alert
82	He's dead captain	131	Sulu: All hands to battle stations
83	He's dead Jim	132	Sulu: All phaser banks fire
84	Most illogical	133	Uhura: Klingon vessel acknowledge please
85	You can't argue with a machine	134	Uhura: sensor picking up a Klingon battle cruiser
		135	Chekov: Number two shield is gone
…		136	Kirk: Beam me aboard
		137	Kirk: Beam me up Mr. Spock
104	Small explosion clipped	138	Kirk: How bad is it
105	Small explosion clipped	139	Kirk: Phasers on stun
106	Small explosion clipped	140	Kirk: Prepare to attack
107	Khan!	141	Kirk: We're running out of time
108	Phaser	142	Kirk: Activate tractor beam
109		143	Kirk: We need power

Sounds Pg 3

Arduino	Sound	Arduino	Sound
144	McCoy: I could cure a rainy day	180	
145	McCoy: Fascinating	181	Bridge background
146	McCoy: I'm a doctor not a bricklayer	182	Brass monkeys background
147	McCoy: I'm a doctor not a mechanic	183	Time reverse background
148	McCoy: Most cold-blooded man I've ever known	184	Trouble with Tribbles background
149	Scotty: Alright you lovelies	185	Silvery orbs background
150	Scotty: We'll get more speed out of her	186	Goodbye Mr. Decker background
151	Spock: Any chance these are hallucinations	187	Violent Shakes
152	Spock: I'm sure there is an answer	188	Opening fanfare
153	Spock: Engine systems coming on	189	Kirk's Explosive Reply
154	Spock: Extremely interesting	190	Surprise Attack
155	Spock: Illogical	191	Klingons ST3
156	Spock: Irritating one of your earth emotions	192	Khan's Pets
157	Spock: Live long and prosper	…	
158	Spock: Can you manage warp 7	202	Powerup Chime
159	Spock: This is not a drill	203	{Powerup Tune
160	Scotty: I'll sit on the warp engines and nurse them	204	Coin up Noise
161	Kid Mode: It's a big galaxy, Mr. Scott	205	Coin up Tune
162	Kid Mode: Hello, Computer	206	{Startup Noise
163	Kid Mode: Scotty, energize!	207	Startup Tune
164	Kid Mode: Fully Operable Scotty? Yes sir.	208	
165	Kid Mode: Warp factor one!	209	
166	Kid Mode: Battle Stations!	210	
167	Kid Mode: USS Enterprise calling Klingon vessel. Acknowledge please.	211	10 Noise
		212	100 Noise
…		213	1000 Noise
		214	Saucer Pre-Tune
		215	Saucer Noise
		216	Bally Bonus Tune
		217	Tilt Tune
		218	Game Over Tune

DIP Switches

The original Bally/Stern MPUs had 32 DIP switches for the purpose of customizing the games. Most, if not all, replacement boards have the same switches. The Arduino reads the settings of these switches, and uses them in similar, if not identical fashion. See the explanations and table below for information on individual switches.

No.	Switch
1	Games per coin (or coins per game) for coin chute #1. Switches 1 – 5.
2	"
3	"
4	"
5	"
6	High score to date award setting. Switches 6 – 7.
7	"
8	Sound option (together with switch 32. Ignored by the Arduino).
9	Games per coin (or coins per game) for coin chute #3. Switches 9 – 13.
10	"
11	"
12	"
13	"
14	Assign award level to playfield specials and scoring thresholds. Switches 14 – 15.
15	"
16	Number of balls per game (3 or 5).
17	Maximum credits. Switches 17 – 19.
18	"
19	"
20	Number of credits to be displayed, yes or no.
21	Match Feature, on or off.
22	Hyperspace value starts at 2,000 or 4,000.
23	Bally special remains lit, yes or no.
24	Center target remains lit or alternates.
25	Games per coin (or coins per game) for coin chute #2. Switches 25 – 28.
26	"
27	"
28	"
29	Bally starts at 10,000 or 25,000.
30	Outlane special, both lanes lit or alternate.
31	Flipper feed lanes, both lit for hyperspace or alternate.
32	Sound option (together with switch 8. Ignored by the Arduino).

Switches 01 – 05, 09 – 13, and 25 – 28: Payment for Games

The first five switches are used together to set the number of credits per coin, or coins per credit, for coins dropped into coin chute #1. The original machines set up 32 different payment schemes. Only one is not reproduced (rows 1 and 2 in the chart below. Both were 3 credits for two coins, with one credit on the first coin and two on the second, unless you score points between the first and second coin, in which case you only get one. It really didn't seem worth the effort!)

Switches 09 – 13 are set up the same, setting the credits per coin for coin chute #3.

Chute			**Switches**			**Credits / Coin**
#1	**5**	**4**	**3**	**2**	**1**	**(No credits until**
#3	**13**	**12**	**11**	**10**	**9**	**2nd coin is dropped)**
1	OFF	OFF	OFF	OFF	OFF	3 Credits / 2 Coins
2	OFF	OFF	OFF	OFF	ON	3 Credits / 2 Coins
3	OFF	OFF	OFF	ON	OFF	1 Credit / 1 Coin
4	OFF	OFF	OFF	ON	ON	1 Credits / 2 Coins
5	OFF	OFF	ON	OFF	OFF	2 Credits / 1 Coin
6	OFF	OFF	ON	OFF	ON	2 Credits / 2 Coins
7	OFF	OFF	ON	ON	OFF	3 Credits / 1 Coin
8	OFF	OFF	ON	ON	ON	3 Credits / 2 Coins
9	OFF	ON	OFF	OFF	OFF	4 Credits / 1 Coin
10	OFF	ON	OFF	OFF	ON	4 Credits / 2 Coins
11	OFF	ON	OFF	ON	OFF	5 Credits / 1 Coin
12	OFF	ON	OFF	ON	ON	5 Credits / 2 Coins
13	OFF	ON	ON	OFF	OFF	6 Credits / 1 Coin
14	OFF	ON	ON	OFF	ON	6 Credits / 2 Coins
15	OFF	ON	ON	ON	OFF	7 Credits / 1 Coin
16	OFF	ON	ON	ON	ON	7 Credits / 2 Coins
17	ON	OFF	OFF	OFF	OFF	8 Credits / 1 Coin
18	ON	OFF	OFF	OFF	ON	8 Credits / 2 Coins
19	ON	OFF	OFF	ON	OFF	9 Credits / 1 Coin
20	ON	OFF	OFF	ON	ON	9 Credits / 2 Coins
21	ON	OFF	ON	OFF	OFF	10 Credits / 1 Coin
22	ON	OFF	ON	OFF	ON	10 Credits / 2 Coins
23	ON	OFF	ON	ON	OFF	11 Credits / 1 Coin
24	ON	OFF	ON	ON	ON	11 Credits / 2 Coins
25	ON	ON	OFF	OFF	OFF	12 Credits / 1 Coin
26	ON	ON	OFF	OFF	ON	12 Credits / 2 Coins
27	ON	ON	OFF	ON	OFF	13 Credits / 1 Coin
28	ON	ON	OFF	ON	ON	13 Credits / 2 Coins
29	ON	ON	ON	OFF	OFF	14 Credits / 1 Coin
30	ON	ON	ON	OFF	ON	14 Credits / 2 Coins
31	ON	ON	ON	ON	OFF	15 Credits / 1 Coin
32	ON	ON	ON	ON	ON	15 Credits / 2 Coins

Switches 25 – 28 are set up slightly differently, for coin chute #2.

Chute #2	Switches 28	27	26	25	Credits / Coin
1	OFF	OFF	OFF	OFF	Same as chute #1
2	OFF	OFF	OFF	ON	1 Credit / 1 Coin
3	OFF	OFF	ON	OFF	2 Credits / 1 Coin
4	OFF	OFF	ON	ON	3 Credits / 1 Coin
5	OFF	ON	OFF	OFF	4 Credits / 1 Coin
6	OFF	ON	OFF	ON	5 Credits / 1 Coin
7	OFF	ON	ON	OFF	6 Credits / 1 Coin
8	OFF	ON	ON	ON	7 Credits / 1 Coin
9	ON	OFF	OFF	OFF	8 Credits / 1 Coin
10	ON	OFF	OFF	ON	9 Credits / 1 Coin
11	ON	OFF	ON	OFF	10 Credits / 1 Coin
12	ON	OFF	ON	ON	11 Credits / 1 Coin
13	ON	ON	OFF	OFF	12 Credits / 1 Coin
14	ON	ON	OFF	ON	13 Credits / 1 Coin
15	ON	ON	ON	OFF	14 Credits / 1 Coin
16	ON	ON	ON	ON	15 Credits / 1 Coin

Switches 06 and 07: High Score Award

Award for Beating High Score	Switch 7	Switch 6
No Award	OFF	OFF
One Credit	OFF	ON
Two Credits	ON	OFF
Three Credits	ON	ON

Switch 08: Sound Options

Switches 8 and 32 originally worked together to produce a cacophony of electronic tunes, buzzes, chimes, and knockers. These two switches are ignored.

Switches 14 and 15: Playfield and Scoring Threshold Special Awards

Switch 14	**ON**	**OFF**	**OFF**
Switch 15	**ON**	**ON**	**OFF**
Outlane Special	Replay	X-Ball*	50,000
Drop Target Special	Replay	X-Ball*	50,000
B-A-L-L-Y Special	Replay	X-Ball*	50,000
Hyperspace Special	X-Ball	X-Ball**	25,000
Scoring Thresholds	Replay	X-Ball**	NO AWARD

* 50,000 awarded if Same Player Shoots Again already lit
** 25,000 awarded if Same Player Shoots Again already lit

Switch 16: Balls Per Game

Liberal	ON	5 balls per game
Conservative	OFF	3 balls per game

Switches 17 - 19: Maximum credits allowed (as per original Bally manual)

Maximum Credits	Switches 19	18	17
5	OFF	OFF	OFF
10	OFF	OFF	ON
15	OFF	ON	OFF
20	OFF	ON	ON
25	ON	OFF	OFF
30	ON	OFF	ON
35	ON	ON	OFF
40	ON	ON	ON

Switch 20: Credits displayed, YES or NO

Switch 21: End of game match feature, ON or OFF

Switch 22: Hyperspace Adjustment

Liberal	ON	Hyperspace begins at 4,000
Conservative	OFF	Hyperspace begins at 2,000

Switch 23: Bally Special Adjustment

Liberal	ON	Bally Special remains lit
Conservative	OFF	Bally Special awarded once

Switch 24: Center Target Adjustment

Liberal	ON	Center target remains lit (3,000 points)
Conservative	OFF	Center target alternates (300 / 3,000 points)

Switch 29: Bally 10,000 – 25,000 Adjustment

Liberal	ON	Bally begins at 25,000
Conservative	OFF	Bally begins at 10,000

Switch 30: Outlane Special Adjustment

Liberal	ON	Both lanes lite for special
Conservative	OFF	Special alternates from side to side

Switch 31: Flipper Feed Lanes Adjustment

Liberal	ON	Both lanes lite for Hyperspace value
Conservative	OFF	Hyperspace award alternates between lanes

Switch 32: Sound Options

Switches 8 and 32 originally worked together to produce a cacophony of electronic tunes, buzzes, chimes, and knockers. These two switches are ignored.

Operator Game Adjustments

There is a section near the top of the main program titled "Operator Game Adjustments" (following the version history), which can be used to adjust features of the game. It includes a number of very important options, such as indicating to your Arduino what music files you have. If you are recompiling the software to your Arduino, you will want to look at these potential changes:

I_HAVE_ADDED_MUSIC_FILES: Change this setting to either true or false, depending on whether or not you have added music to the game. This software does not provide the music for copyright reasons, although you can easily find the music and add it to your game. See the section titled "Adding Music Files to the WAV Trigger" below for additional information.

KOBAYASHI_MARU_TIME: You must disable a Klingon every 15 seconds in the Kobayashi Maru test. This setting allows you to change this to a higher or lower number.

NACELLE_PATTERN: This sets the nacelle light pattern for the majority of the game. There are currently 31 possible patterns, but you likely want pattern 0, 1, or 2 (no lights, bright lights, or dim lights). This also assumes you have nacelle lights. See the Notes section below, "Adding Nacelle Lights" for more information.

BALL_SEARCH_TILT: If a ball search is unsuccessful, the game will turn off the tilt mechanism, allowing the player to nudge the machine in order to free a stuck ball. Change this setting to 1 to allow this feature, or 0 to disallow it.

ATTRACT_SPEECH_TIMER: Time between attract mode callouts. Usually 300000 ms, or 5 minutes.

MAX_TILT_WARNINGS: Usually 1. Set higher if you wish to allow some number of tilts before ending a ball.

VERSION_NUMBER: The program expects a version number in the format yyyy.mm, representing the year and month of your changes. This version number is displayed when the pinball is turned on. If you are modifying the program, you may want to copy the software to a new directory named STyyyypmm, and rename the two files that use this naming format. Then change this version number in the software to make clear which version is running.

DEBUG_MODE: There are a number of programmable features available to assist you in identifying and possibly eliminating some common pinball issues. Currently there are three debug modes, available by setting DEBUG_MODE to one of the following values:

0. Set to zero to indicate no debug messages are required (regular game play).
1. Monitor the switches as they are hit. The most recent nine switches are scrolled through the player 2, 3, and 4 displays.

2. Display the number of times per second the switches are monitored. Player 4 display is used. The result is highly game and activity-specific. The result for the Star Trek is usually around 1000 – 1200, although for more complex games the result is often in the range of about 180 to 200 times a second.

By playing a single player game with debug mode 1 set, you can monitor the switches being activated in real time. This can be very useful if you are experiencing switch issues. Note that it must be a single player game, as the player 2, 3, and 4 displays are used to display the switch values. Each 6-digit display will show three two-digit switch numbers. The most recent switch is displayed in the last two digits of Player 4, and will scroll left and up as new switches are hit.

Pressing and holding the game credit button will stop the monitoring of new switch hits, for as long as you hold the button. This allows you to stop the numbers from scrolling away, so you can review them more closely.

Adding Music Files to the WAV Trigger

The ST2026, unfortunately, does not and cannot provide music from the Star Trek show or movies. Background sounds are therefore all set up to play Star Trek bridge sounds. However, adding the music is easy. Watch the following YouTube video for step-by-step instructions on purchasing and adding Star Trek music to your pinball:

https://youtu.be/CJBlf6C11SE?si=8CX0rF0vCIO9eKpC

The files I have used for my own pinball include:

Music	Expected File Name*	Plays During…
Bridge Background Sounds	0181_bridge_background.wav (included)	Normal Gameplay
Star Trek: Brass Monkeys	0182_brassmonkeys.wav	Normal Gameplay
Star Trek: Time Reverse / Future Risk	0183_timereversefuturerisk.wav	Normal Gameplay
Star Trek: Trouble with Tribbles Suite	0184_troublewithtribbles.wav	Normal Gameplay
Star Trek: Silvery Orbs	0185_silveryorbs.wav	Normal Gameplay
Star Trek: Goodbye Mr. Decker	0186_goodbyemrdecker.wav	Normal Gameplay
Star Trek: Violent Shakes	0187_Violent Shakes.wav	Normal Gameplay
Wrath of Khan: Kirk's Explosive Reply	0189_kirksexplosivereply.wav	Challenge 1
Wrath of Khan: Surprise Attack	0190_surpriseattack.wav	Challenge 2
Search for Spock: Klingons	0191_KlingonST3.wav	Challenge 3 (Running)
Wrath of Khan: Khan's Pets	0192_KhansPets.wav	Challenge 3 (Hiding)

This music, and many other pieces from the TV show and movies, is available online for as little as a dollar per song. Once purchased, you will very likely have to convert the files to WAV format. The easiest way I have found to do this is to use the WavePad Audio Editor, available for free at: https://www.nch.com.au/wavepad/

* The number is important, indicating when the music is played. However, the rest of the name can be modified.

Just edit the files, then click “Save File As” and select wav file format.

Once you have the music in wav files (using the numbered naming convention above!), you need to make sure they have a sample rate of 44100Hz, encoding with signed 16-bit PCM, and all metadata removed. This sounds very technical, but all you really need to do is download the free Audacity audio editor from: https://sourceforge.net/projects/audacity/

Use this simple tool to open your files and export them (i.e., save them), filling in the given prompts with the requirements listed above (44100 Hz, 16-bit, no metadata). This video may also help:

https://youtu.be/G9YgYAGwlz0?si=0cHR0Hg22QXuNT9y&t=197

IMPORTANT: Be sure to remove all metadata from every wav file! Before exporting, be sure to click on the “Edit Metadata” button, click “Clear” to delete all the artist/track/etc. data, and click OK to be sure this information is gone! The WAV Trigger cannot read the files if any of this data is present.

Now just copy these files onto the micro-SD card for your WAV Trigger board.

Finally, you may need to make a minor change to the software. Go to the Operator Game Adjustments and set the value I_HAVE_ADDED_MUSIC_FILES to true. See the section “Operator Game Adjustments” above for more information.

Now recompile the program to your Arduino!

Other Programmable Changes*

SwitchDebounce[]

This variable can be found near the top of the main program, shortly following the Operator Game Adjustments. It contains 40 rows, one for each switch, and two columns. It is used in monitoring the switches for multiple hits. Occasionally, a switch can develop a "bounce", where it registers two or more times for a single hit by a pinball. By setting a value from 0 to 255 in the second column, the pinball will ignore any hits on that switch for that number of milliseconds following an initial hit.

For example, I was having trouble with the left outlane rollover (switch 34) registering multiple times whenever it was hit. To fix this I set column 2 of row 34 to 250. So now, after the switch is hit it will not register another hit for at least 250 milliseconds.

The Switch Bounce Test can be used to test switches for bounce. See the Switch Bounce Test documentation in the Self-Test section of this document. This test will also tell you the time between double-hits, which can be used as a minimum value in setting the second column of this variable.

ResetHits[]

This variable can also be found near the top of the main program, right after the SwitchDebounce[] variable. It contains 14 rows, one for each solenoid, and 2 columns. It is used to eliminate switches from activating due to vibration from solenoids firing. Similar to SwitchDebounce[], a value from zero to 255 can be set in the second column, causing the pinball to ignore hits to specific switches for the indicated number of milliseconds following a solenoid firing.

To make this work, you also need to indicate which switches are to be ignored following which solenoid firing. This is done by adding code to the function ResetHitFix(). There are examples of the required code in the function already, which you can use to model your own.

For example, suppose that resetting the drop targets is causing the right outlane switch to activate. You've tried cleaning and re-gapping the switch but can't seem to get it to stop. To fix this you can set row 13 (SO_DTARGET_RESET), column 2 of ResetHits[] to 250. Then add the following code to ResetHitFix():

```
case SW_RIGHT_OUTLANE:
    if (CurrentTime < ResetHits[SO_DTARGET_RESET].start) return false;
    if (CurrentTime - ResetHits[SO_DTARGET_RESET].start
            < ResetHits[SO_DTARGET_RESET].wait) return true;
    return false;
```

* This section requires some programming skill. If you do not feel qualified, maybe skip this section! Always make a copy of the software before attempting to make changes.

This says, if the right outlane target is hit, and a drop target reset has occurred, check how long it has been since the drop target fired. If less that 250 milliseconds ago, the switch hit is to be ignored (return true).

Coding has already been added to prevent any drop target solenoid from setting off its own drop target switches. This can be used as an example to model your own code.

Setting the Wait Time

Both SwitchDebounce[] and ResetHits[] depend on you setting a “wait” time. This is the time between events during which the pinball will ignore specific switches. The wait time must be longer than the time it takes for vibration to set off the second switch hit, but shorter than the time it can take for a switch to be hit legitimately, with a maximum of 255ms.

250ms, or a quarter of a second, is generally a good choice.

The time between a solenoid firing, and a switch being activated by vibration, can be determined in the Solenoid self-test. The time taken for a switch bounce can similarly be determined in the Switch Bounce self-test. The results of these tests would be the absolute minimum value you should use for the wait time, although you likely want something higher.

For example, I had a drop target that would frequently set off a target switch on the other side of the playfield. It would be impossible to activate the drop target solenoid and then the target switch in less than several seconds, much longer than the maximum 255ms. The solenoid test showed that it took about 133ms between the drop target reset until the switch would activate. The wait time could then be an absolute minimum of 133, up to a maximum of 255. I therefore set the wait time to 250, allowing extra time over and above the absolute minimum.

Generally, a longer time is better, since the wait time could be longer than that given by the Solenoid or Switch Bounce test. Where it could be more difficult is when the solenoid and the target switch are close together. For example, a pop bumper that can shoot the ball directly into a switch. If the pop bumper is sometimes setting off the switch without hitting it, you will need to determine not only the minimum time from the solenoid test, but also a maximum time based on how long it can take the ball to travel from the pop bumper to the switch. If this is less than a quarter second, you may need to do some careful timing calculations.

Sound Changes

This program was not written with replacing the sounds in mind! However, it is possible. Just, not easy. The following suggestions should help. Search the internet for “Robertsonics WAV Trigger” for additional requirements and suggestions when creating your own sound files. The Audacity audio editor is recommended. 44100, 16-bit, stereo, with all metadata deleted, is required.

Sound files can be changed on the WAV Trigger’s micro-SD card. Notice that every file begins with a 4-digit number between 0001 and 0255. The WAV Trigger looks at this number, and ignores the remainder of the file name. By replacing any sound file with a file of your own that begins with the same number, the game will play your file instead. For example, to change the background music, you could make up your own files 0181 – 0190. You should also review the variable SoundTimings, near the bottom of file ST2026p04.h. This records the sound type (background, voice, or sound effect) and length (in 10ths of a second) for each file (background music files should have length set to zero). Note that files with sound type zero will not play!

Changing more than a very few files this way could get tedious. If you want to change most or all of the files, I recommend that you create a new directory on your computer. Create all your new sound files in this directory. Assign them all numbers from 0001 to 0255, and add these numbers to the beginning of each sound file. Don’t worry about matching my numbers.

You will then need to rewrite the SoundTimings variable (near the end of ST2026p04.h), assigning sound types and lengths for all your sounds.

In ST2026p04.h, below the SoundTimings variable, are a number of variables set up for the purpose of assigning sounds to events in the game. Some of these are lists of sound file numbers, and the event will choose from the list randomly.

For example: VoiceTilt[4] = {3, 27, 33, 76};

This variable is used to select one of three voices to be played when the game is tilted. The first number (3) must be the number of sound files in the group, followed by that many sound files (in this case, 27, 33, and 76).

Others are single values, for example: VoiceBallSave = 125;

This indicates that sound file 125 will be played when ball save is applied (“Try again”).

Once you have made up your directory of sound files, go through these variables, deleting my numbers where you want and replacing them with yours. Save the old files from the micro-SD card in another directory, and replace them with the new files from your directory. Finally, recompile the program to your Arduino. I recommend you keep a copy of your new ST2026p04.h file with the sound files, as they are required to work together. A directory of sound files played with an incorrect SoundTimings or assignment variables will not work properly.

I warned you it wasn’t going to be easy!

Alternative Game Settings

Kids' Mode is described above, and is meant to allow you to convert the game to easy settings, primarily for kids to play. However, another use of this might be to allow players with different setting preferences to share a pinball machine. For example, maybe you like a five-ball game, but your brother insists on three. In this case, you might not want ALL the settings at easy levels, only the number of balls.

This would be easy to do. There is a function KidSettings() at the end of the main program file. By commenting out or rewriting the lines in this function, Kids' Mode can be adapted to the alternative game settings you prefer.

You can then follow the instructions for "Game Mode" to get into your alternative setting:

Game mode is selected at the start of a new game by pressing and holding the game button, either during attract mode or before player 1 begins play. After holding the button for one second, the credit window will begin scrolling the values 1, 2, and 3. Release the game button when 1 is displayed for regular play, or when 2 is showing to play your alternate game settings.

Notes

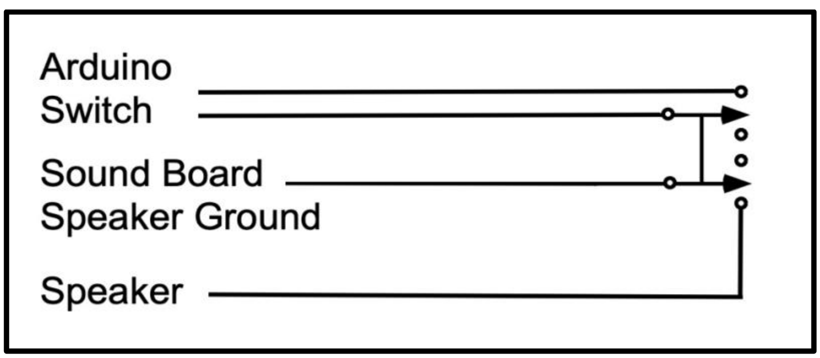
DIP Switches: The DIP switches for the Arduino have been set up to be as identical to the switches for the regular game as possible. So simply installing the Arduino in your pinball should have it run according to the rules you are familiar with. See the original Bally Star Trek manual for more information on the DIP switches, and on the self-tests and game settings.

A new test has been added to the self-tests, to display and change the DIP switch settings easily, but temporarily. See the Self-Test section above for more information.

Self-Test Settings: The original self-test settings, stored on your Star Trek MPU board, are not used by the Arduino. The first thing you should do after installing the Arduino and before playing a game is to enter Self-Test, and update all the game settings to the values you want.

Switch to Original: It is possible to add a switch to the Arduino, on long wires running from the 2-pin connector labeled "Switch", and then out the air vents at the back, so that you will be able to easily switch back and forth between the new and old rules. But don't bother. You are never going to want to play the old rules again!

If you do want to be able to play the old rules, keep in mind that the WAV Trigger board will not play the old sounds. You will need to set up another switch to control where the sound is directed. I would recommend a double-pole, double-throw switch, which allows you to control both with a single switch as below:



When the switch is up, the Arduino switch is connected and the sound board speaker is disconnected, so game and sound are controlled by the Arduino. When the switch is down, The Arduino switch is disconnected and the sound board is connected to the pinball's original speaker, so the original game is running and playing the original sound through the original speaker.

End Game: If you press the game button during ball 2 or later of a regular game, the game will end and you will be sent to Attract Mode. I'm not sure why you would want that, but it is apparently a feature of the original Bally pinball and so it is recreated here.

Adding Nacelle Lights: The Bally Star Trek has an image of the U.S.S. Enterprise in the middle of its playfield. This iconic ship has two tubes, one on either side, which are displayed prominently in this image. These are known as the “nacelles”, from a French term indicating a “housing or tank on the outside of a vessel”. In the show, the front of these tubes glow brightly.

If you have replaced your playfield with one from Classic Playfield Reproductions (CPR), you will find that these nacelles, which were originally painted on the playfield, have been replaced with plastic playfield inserts that can be lit from underneath.

There are a number of ways you can set up the lighting of the nacelles. The easiest way is to add a couple of lamps, and connect them to the nearest GI lights. This would ensure they were both brightly lit at all times.

Another alternative is, you could connect them to the two switched illumination lines that run right past them. Then connect the return line from the left nacelle light to J3-15 on the Lamp Driver Board, and the right nacelle light to J1-23 on the Lamp Driver Board. This makes lamp #11 the left nacelle, and lamp #12 the right nacelle. See the following video for more information:

https://youtu.be/T1R3IDWx6xI?si=QDMfOJ3fUb8vKtip&t=645

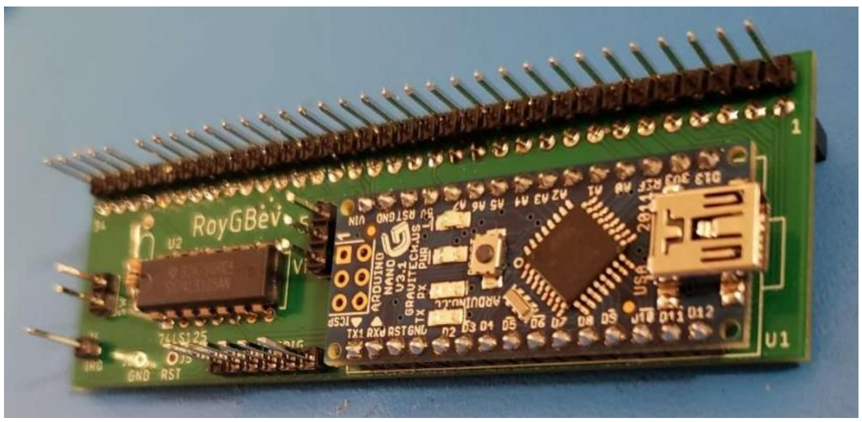
The Arduino software is currently written to recognize these nacelle lights if set up this way. Currently, very little is done with these inserts (hey, I only wired them up on Thursday!), although there may be more in the future. See the section, “Operator Game Adjustments: Nacelle_Pattern” above for information on setting up these lights to be used throughout the game.

Run the light test, press the game button until one of the nacelles comes on, and then double-click the game button to start a review of the nacelle patterns. Then press the game button repeatedly, or just hold it, to step through all the possible nacelle patterns. There are currently 31, including all combinations of five brightness / flicker levels applied to both lights.

Arduino Installation Instructions [*]

Part Description	Qty
Bally/Stern Arduino adapter PCB	1
Push-release IRQ wire harness	1
(Optional) .wav trigger sound board	1
(Optional) 6-pin ribbon cable 1	1

The Arduino adapter in your kit will be one of two types: The REV 1 board (Figure 1), or the REV 3 board (Figure 2).



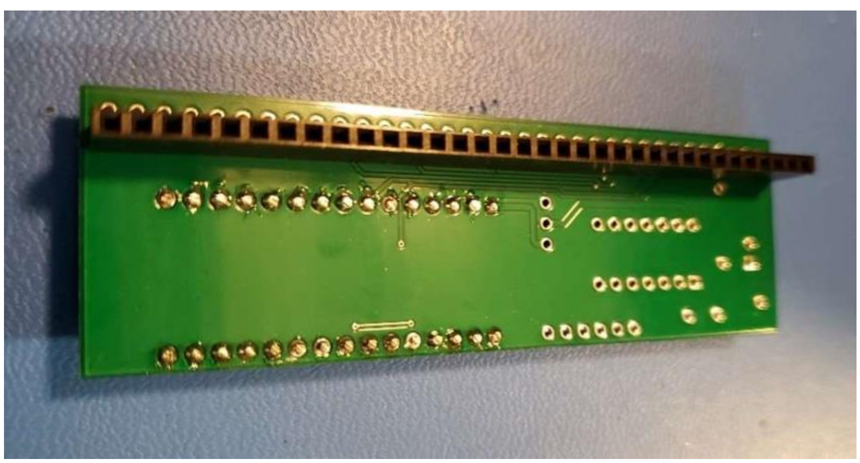


Figure 1

[*] These are the complete installation notes provided with an Arduino Mega 2560 REV 3. They are unedited, and so may contain extra information not needed in installing your Star Trek Arduino. It also includes full instructions for installing and connecting a WAV Trigger sound board. Additional notes of my own are included at the end.

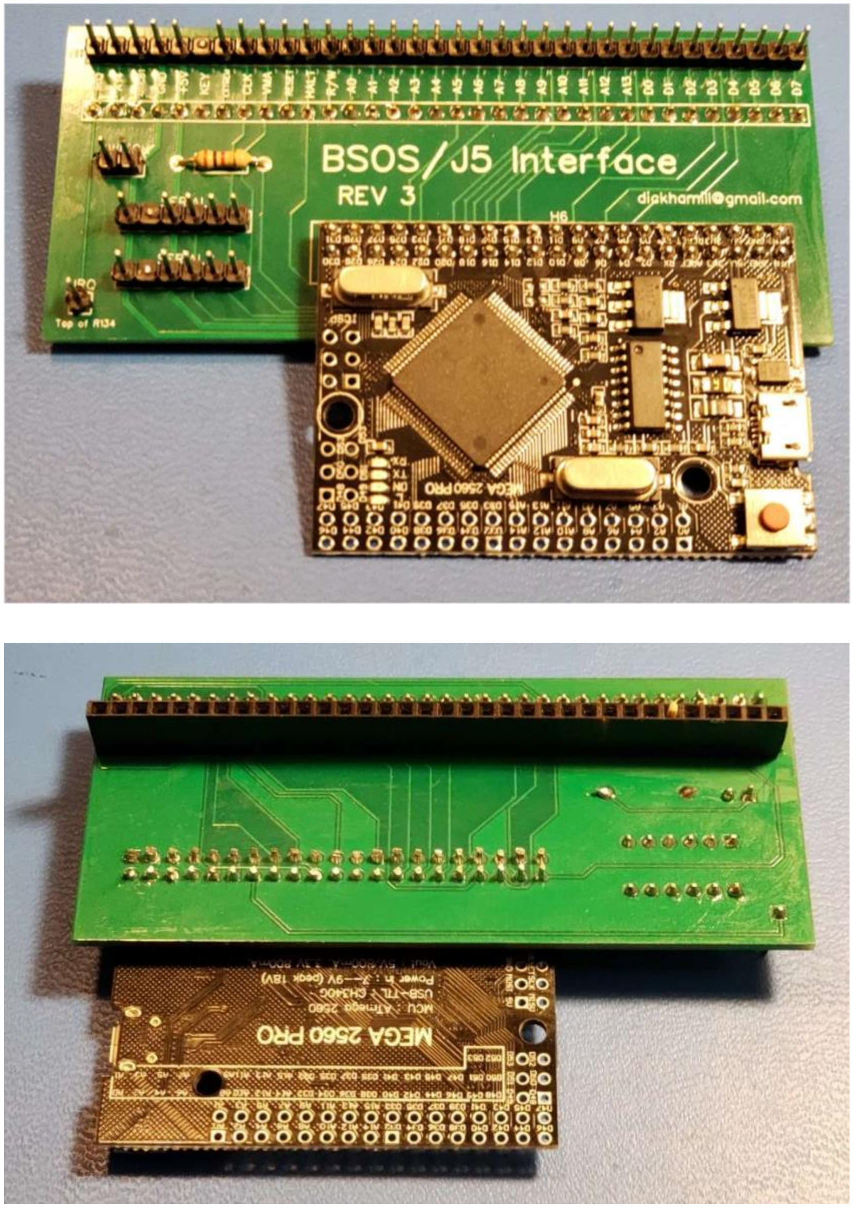


Figure 2

IMPORTANT: TURN OFF POWER TO THE GAME BEFORE INSTALLING THE ARDUINO ADAPTER.

1. Verify that the game's MPU board is in good working condition. Verify that the game powers up and works.

2. If the game has a ribbon cable connecting a sound board to J5 on the MPU, note the orientation of the cable, then remove it.

3. Verify that the MPU connector J5 is in good condition, with good solder joints. Also verify that key pin 29 of J5 has been removed. J5 pins are numbered from pin 1 on the right.

4. If your kit uses the REV 1 Arduino board, install a 0.1” jumper between the Vin and middle pins of the 3-pin header, as shown in Figure 3. The pin labeled 5V should be unconnected.

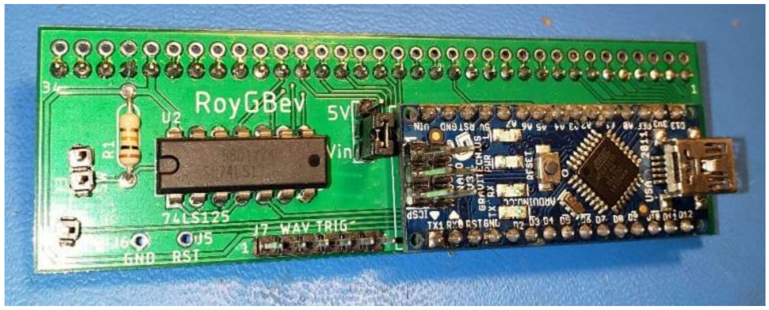


Figure 3

5. The 34-pin female connector on the bottom of the adapter will be plugged into J5 on the game’s MPU board. If you are using an Alltek or Stern MPU-200 board, J5 will have 34 pins. On other boards, J5 may have 32 or 33 pins. In all cases, pin 1 on the adapter will align with pin 1 on J5, and the blocking pin at pin 29 of the adapter will align with the missing pin 29 on J5.

6. Line up pin 1 and carefully install the adapter board onto connector J5. If the connector won’t go, don’t force it. Verify that pin 1 is lined up, and that there are no bent pins preventing engagement. Push the adapter down onto J5 until it is fully installed. Figure 4 shows an adapter installed on an MPU board.

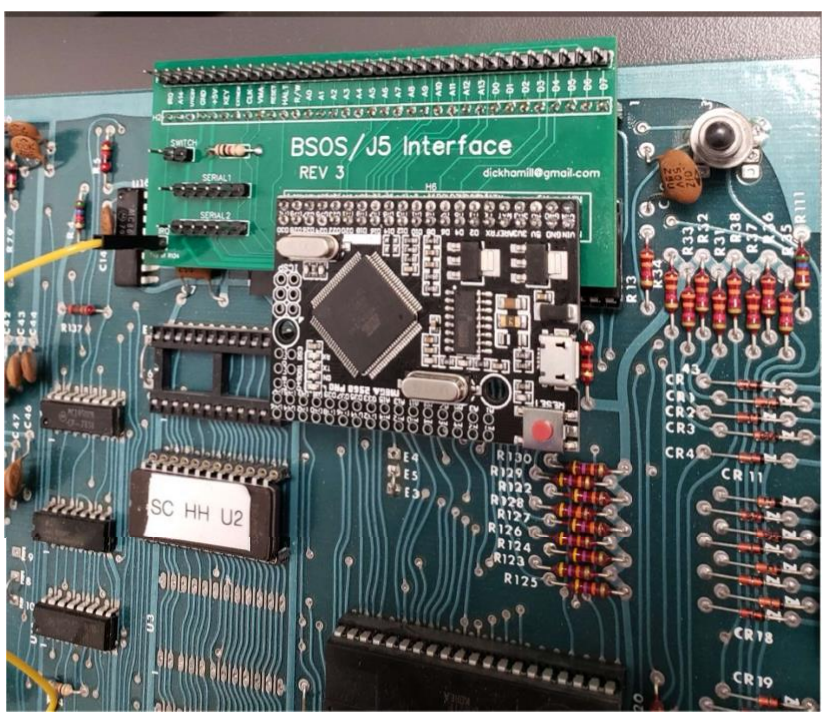


Figure 4

7. If a sound board ribbon cable was removed from J5 in step 2, install the cable onto the 34-pin connector on the Arduino adapter.

8. If your MPU board is a Bally AS-2518-17, Bally AS-2518-35, or Stern MPU-100, the push-release IRQ wire harness will need to be installed. To install:

   a) Push the single pin female connector onto the pin labeled IRQ on the Arduino adapter.
   b) Connect the push-release probe to the top lead of R134 on the MPU board. Figure 5 shows this in detail.

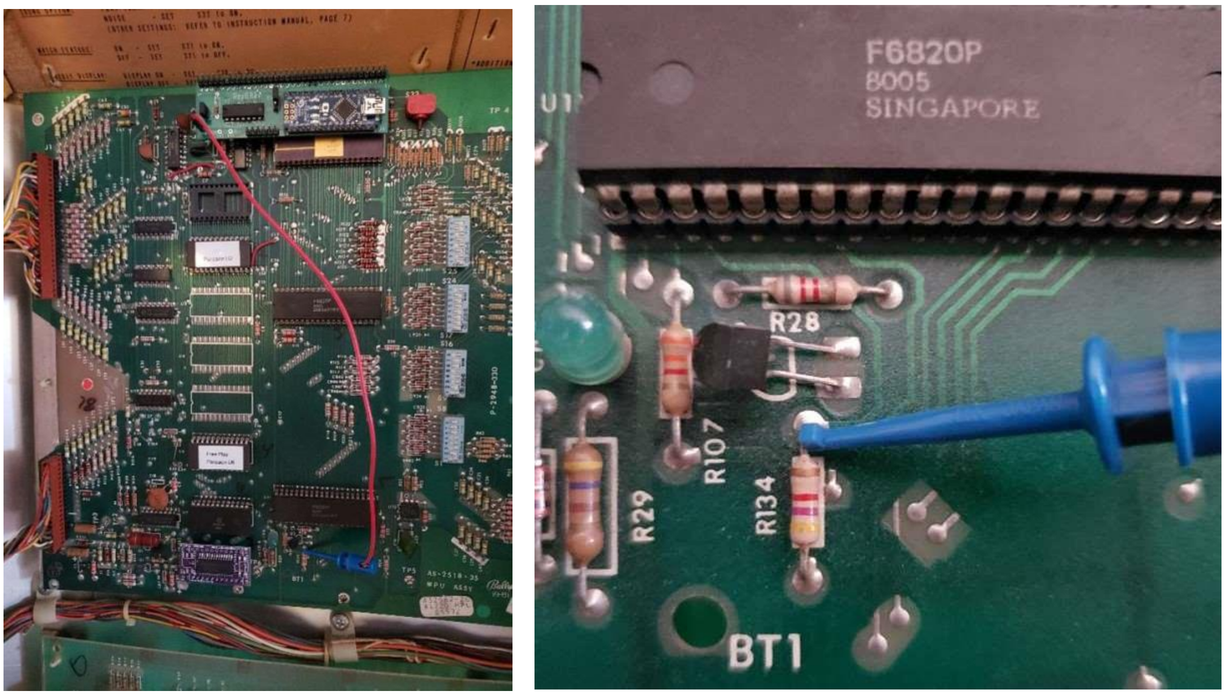


Figure 5

9. The two-pin header labeled SWITCH on the adapter board should not have a jumper installed. Verify this, and remove the jumper if it is installed.

10. At this point the game should be ready to power up. Watch the diagnostic LED on the MPU board while powering up the game. The game should power up normally, to the original 6800 game program, and into attract mode. If it does not, turn the game off and check the installation. If a sound board ribbon cable was attached to the Arduino adapter's 34-pin male header, unplug the cable, then power up the game again. If it powers up normally, there is an issue with the sound board ribbon cable installation.

11. Turn off the game, then install a jumper between the two pins of the connector labeled SWITCH. IMPORTANT: THE GAME MUST BE TURNED OFF BEFORE THE SWITCH JUMPER IS INSTALLED OR REMOVED.

12. Turn on the game. It should boot up to the new Arduino code. Check that the displays show digits, and that the game enters attract mode.

13. If the game does not boot to the Arduino software version, turn off the game and check that the IRQ wire harness is properly installed.

14. If desired, a remote switch may be connected to the pins of the SWITCH connector, allowing the user to select between standard and Arduino versions. As with the jumper, TURN OFF THE GAME BEFORE CHANGING THE SWITCH SETTING.

15. The new Arduino software may support configuration settings that can be selected using the game's self-test switch inside the coin door. Refer to the user documentation for the Arduino pinball software for details.

16. If the Arduino installation kit includes a Robertsonics .wav trigger sound board, refer to the following steps for installation.

17. TURN OFF POWER to the game before proceeding.

18. Refer to Figure 6 for the next few steps.

19. Verify that the Load/Run switch on the sound board is in the Run position.

20. The sound board has a Micro SD card socket with a memory card installed in it. Confirm that the memory card is installed in the socket.

21. Install one end of the 6-pin ribbon cable to the 6-pin connector labeled SERIAL1 (REV 3 board) or J7 WAV TRIG (REV 1 board). The cable has a black stripe on the connector, designating pin 2. Install the cable with the black stripe lined up with pin 2 (the missing pin) on the 6-pin connector.

22. Connect the loose end of the 6-pin ribbon cable to the male header on the sound board. Note that the ribbon cable's connector has a black stripe designating pin 2 on the connector. Line up the black stripe with pin 2 (the missing pin) on the sound board's connector.

23. The sound board has a piece of foam mounting tape attached to the back. This tape can be used to temporarily mount the sound board. Place the sound board in a location close enough that the 6-pin ribbon cable can easily reach. Refer to Figure 7 for an example. For a more permanent installation, the use of mounting screws and standoffs is recommended.

24. The sound board has a 3.5 mm stereo jack for use with speakers. Connect a pair of wired powered speakers, or other wired device (earbuds / headphones) to the jack.

25. To test that the sound board is working, turn the game on. The sound board should play a power up sound when the Arduino game code boots up.

26. If the sound board does not play a sound at power up, test the sound board by pressing the test button. The sound board should make a sound. If it does, but there is no sound

during gameplay, the game software may not be configured to use the sound board. Refer to the user documentation for the Arduino pinball software for details.

27.

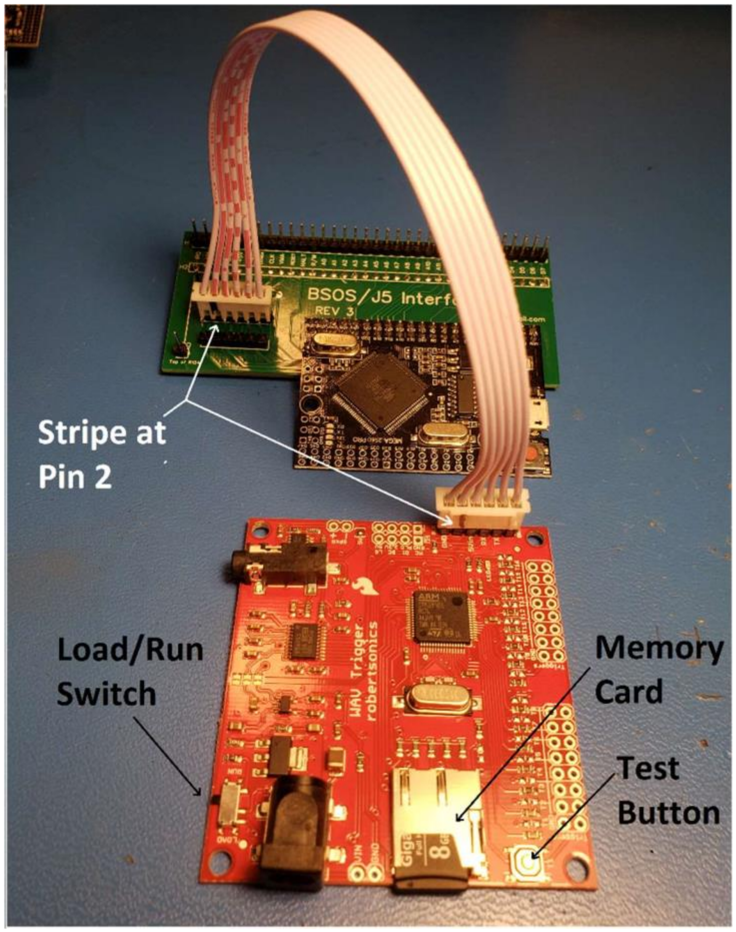


Figure 6

28. Suggested installation: Figures 7 and 8 show an installation for Stern Trident, with powered speakers mounted on top of the game's backbox.

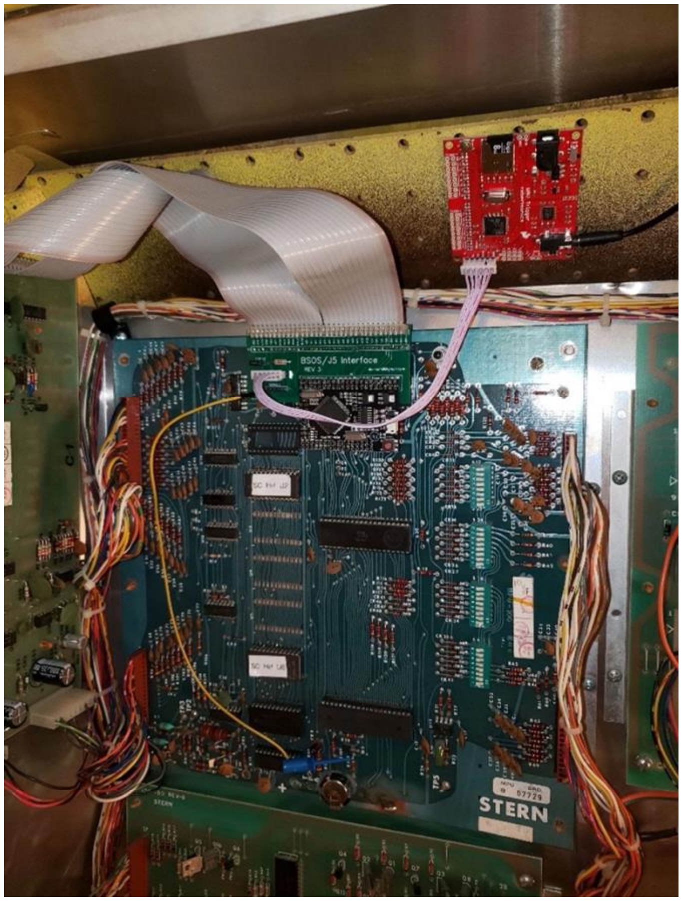


Figure 7

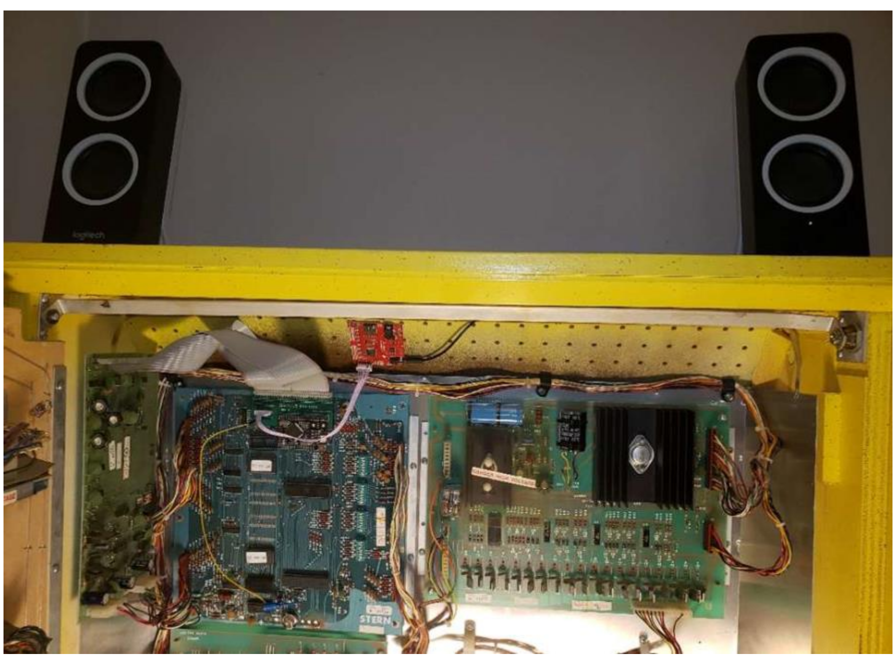


Figure 8

29. Refer any questions regarding this document and installation to roygbevdotcom@gmail.com.

Additional Installation Instructions from Dave's Think Tank

WAV Trigger Installation

If you've purchased an Arduino kit from RoyGBev and it came with a Robertsonics WAV Trigger, then you can follow the instructions he gives you to install it, no problem. However, if you obtained your WAV Trigger separately from another source, then it probably did not come fully prepared. In that case you will likely want to read the following instructions! But, even if you have the prepared kit, there is additional information in this section you may find useful.

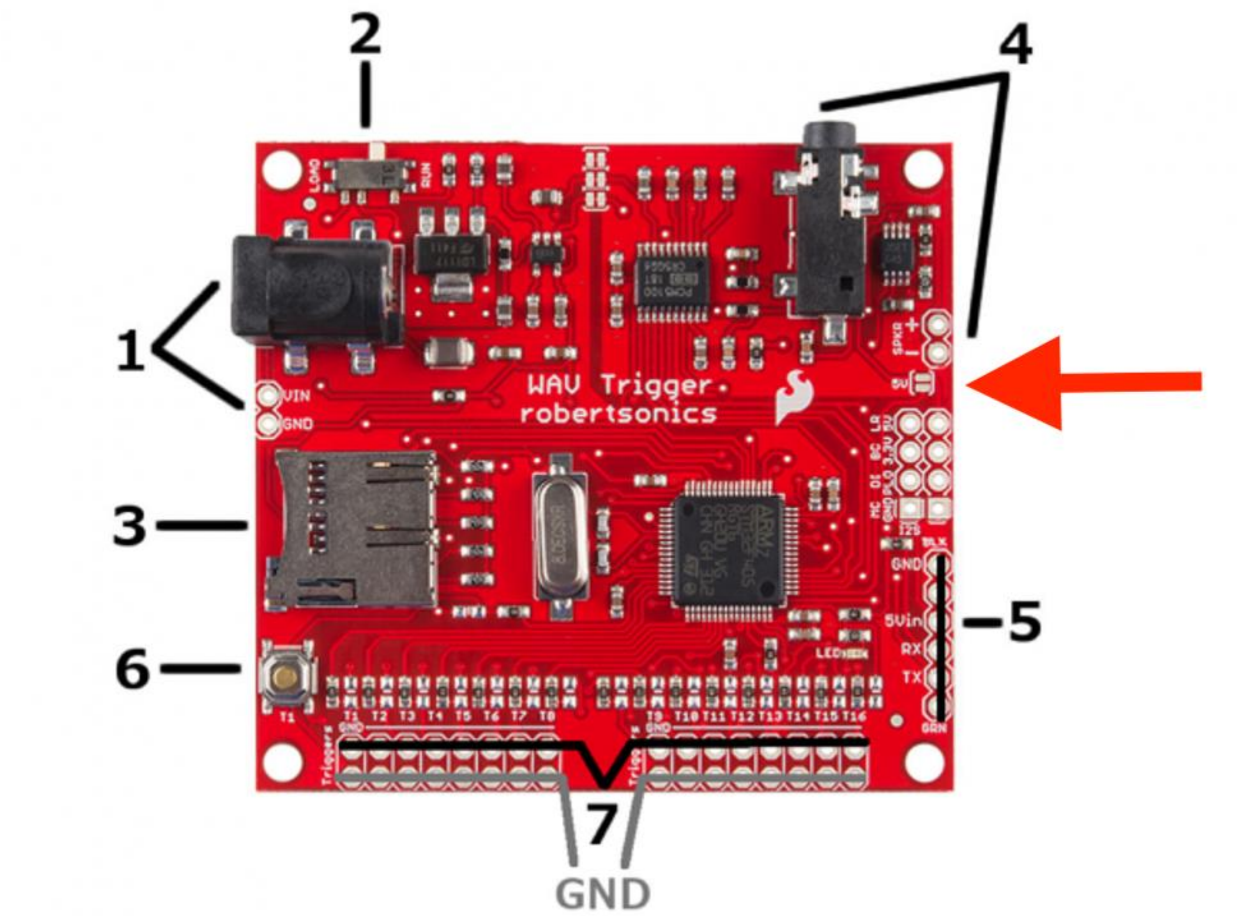


Figure 9

1. The first thing you need to do is purchase a few more items. You are going to need:
    a. A 6-pin ribbon cable. The specs are: Ribbon Cables / IDC Cables Slim Body Single-Row IDC Socket Assemblies / 6 position / Female to Female. This cable should be ten inches or longer.
    b. A 0.1" 6-pin vertical male through-hole header (often sold with 40 pins, which can then be trimmed to six). The ribbon cable attaches to the board at #5 in the diagram above (known as the FTDI port), except the board is sold without pins

soldered here. The required pins have 0.1" spacing, which is the same as used in many pinball board connections, so you may have some of this already.

c. A 32GB or less micro-SD card. Class 10 SD cards with a FAT16 or FAT32 file system format and a 32kB file allocation size are recommended. The card needs to be formatted to 32GB or less, which is plenty for this purpose (generally less than 100 MB). Buy a card 32GB or less, or else format a larger card to 32GB using MS-DOS FAT32.

d. A computer speaker bar. You can pick one up anywhere for under $20, so not expensive. It should have a 3.5mm stereo jack to connect to the WAV Trigger, as well as a USB connector to power the speaker.

e. Four really narrow screws, and eight plastic washers. These are for mounting the WAV Trigger in your backbox. If purchasing, #4 wood screws are what you want. Also, I didn't have tiny washers, so I cut the end rings off some plastic screw anchors and used them.

2. Next, move the switch at #2 on the WAV Trigger to RUN (to the right in the diagram). You will never need it set to LOAD.
3. Place the 6-pin header into the six holes at #5 in the diagram. Turn it over, and solder one pin on the bottom. Turn it over again, and while re-heating the solder, adjust the header to be perfectly vertical. Now finish soldering all six pins. Don't solder all six pins before getting the header upright, or you'll be stuck with it on an angle forever!
4. This is very important, but it is information that is very hard to find! The red arrow in the diagram above is pointing to a "solder bridge", which is two metal pads side by side (labeled 5V). It is sometimes referred to as the "SJ2 solder jumper". The WAV Trigger will not work unless you place a drop of solder across these two traces, connecting them (harder to do than it sounds).
5. Check all your solder connections using your multimeter in continuity mode:
    a. Try connecting the GND pin at #5 to any of the GND pins at #7. They should be connected.
    b. Test to ensure that the 5Vin pin from #5 is connected to the solder bridge.
    c. Test the solder bridge by connecting the 5Vin pin at #5 to the four top pins on the small chip in the #4 corner. I'm not sure which of the pins is the right one, and I haven't been able to find a better place to test!
    d. Try connecting the TX pin at #5 to the large square chip in the lower right quadrant. The correct chip pin is on the right side, about half way up, with a red line connecting it to a tiny hole in the board.
    e. You do not need to test the RX pin, as it is not used by the Arduino. I also do not know where it connects!
    f. Test each of the six pins to make sure they are not accidentally soldered to a neighbouring pin.

6. Your WAV Trigger can now be installed using the four narrow screws through the (very small) holes at the corners of the board. I recommend using plastic washers on top and bottom of the board. On the bottom, to keep the exposed metal connectors on the bottom from touching anything they shouldn't. On top, because the screw heads are likely wide enough to reach exposed parts of the board and you don't want to short them. I now only use three screws, because the hole at the #4 corner is so close to a board component, I am worried about damaging it. I do not recommend attaching the WAV Trigger to the metal sheet on the back of the backbox, as this could easily short out the board if anything touched it. Find a space for it on wood.
7. Copy all sound files to the Micro-SD card (into the root directory!), and insert it into #3, the Micro-SD card reader. Insert carefully until it passes the side spring, then push down to lock the card in place.
8. Your WAV Trigger is now ready to be installed. Connect #5 to the Arduino Serial 1 port with the ribbon cable, and plug the speaker bar in at #4, the 3.5mm jack. Be careful to line up all 6 pins at both ends of the ribbon cable. There is no key on this cable to help orient it! Refer to the photo to make sure you do not get the cable on backward. Note that the red wire in the diagram attaches to the left end of the Arduino, and the right end of the WAV Trigger.

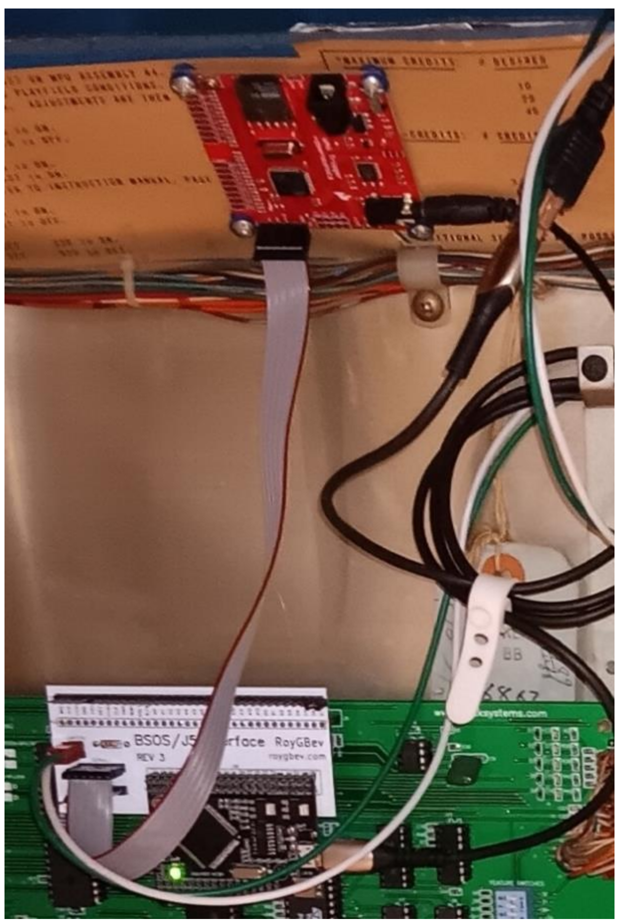


Figure 10

9. I recommend you purchase a stereo computer speaker bar, to be used as the speakers in the above setup. They cost about $20, so not expensive. It should have a 3.5mm stereo jack to connect to the WAV Trigger, as well as a USB connector to power the speaker. Whatever speakers you use, the WAV Trigger does not power the speakers, so external power is required. The USB connector on the computer speaker can be plugged into an old phone charger to complete the setup, or you can tap into the voltage regulator board's 5-volts by following the instructions below.

Powering your Speaker Bar from the Voltage Regulator Board

If you want the speakers to turn off when you turn off the pinball, it is possible to power a small speaker bar (around 1 amp usually) using a 5-volt source in your pinball. This may require some modifications to your voltage regulator board. The method I used is documented below.

Updates to the Voltage Regulator Board

Have you ever noticed on the J3 connector for the voltage regulator board, there is a loop of wire that connects pin 13 to pin 25 (see Figure 11)? This takes 5 volts from one section of the board to another. But if we were to connect those sections directly, this loop could be removed, freeing up two sources of 5 volts which could then be used to power our speaker.

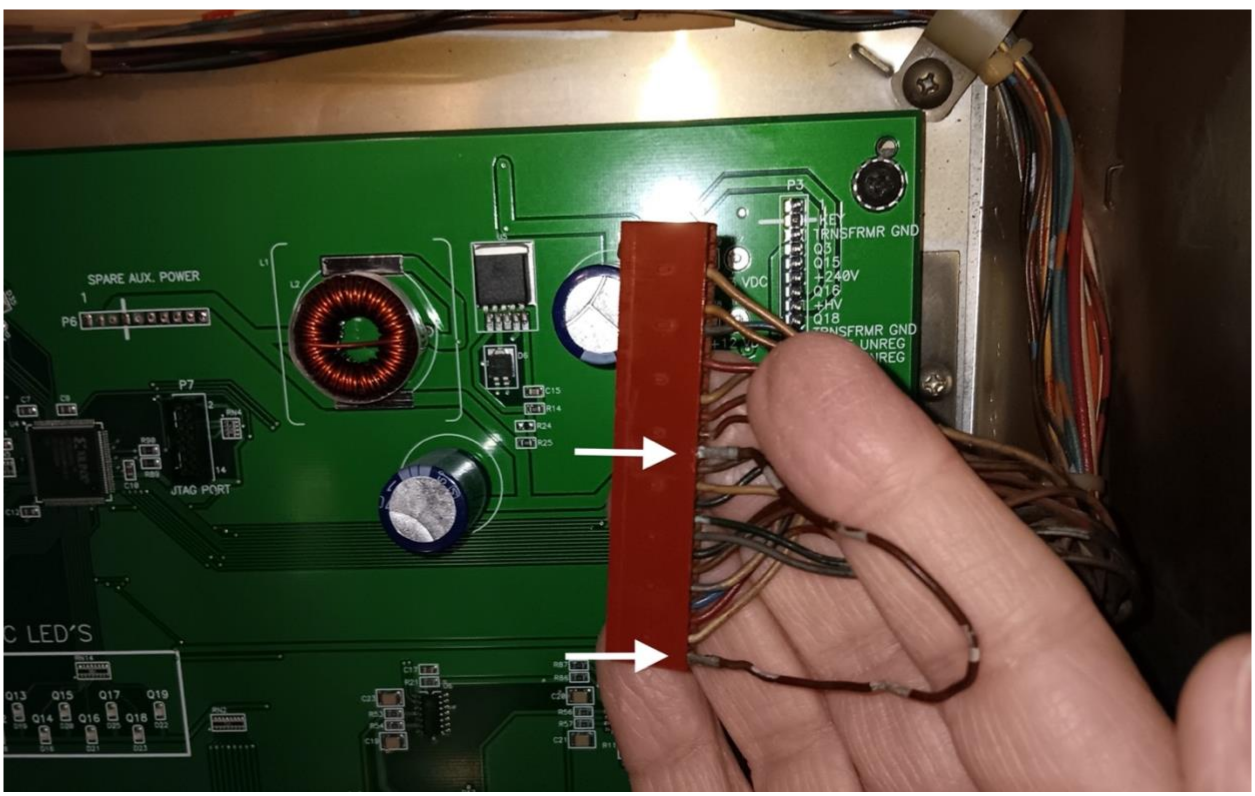


Figure 11

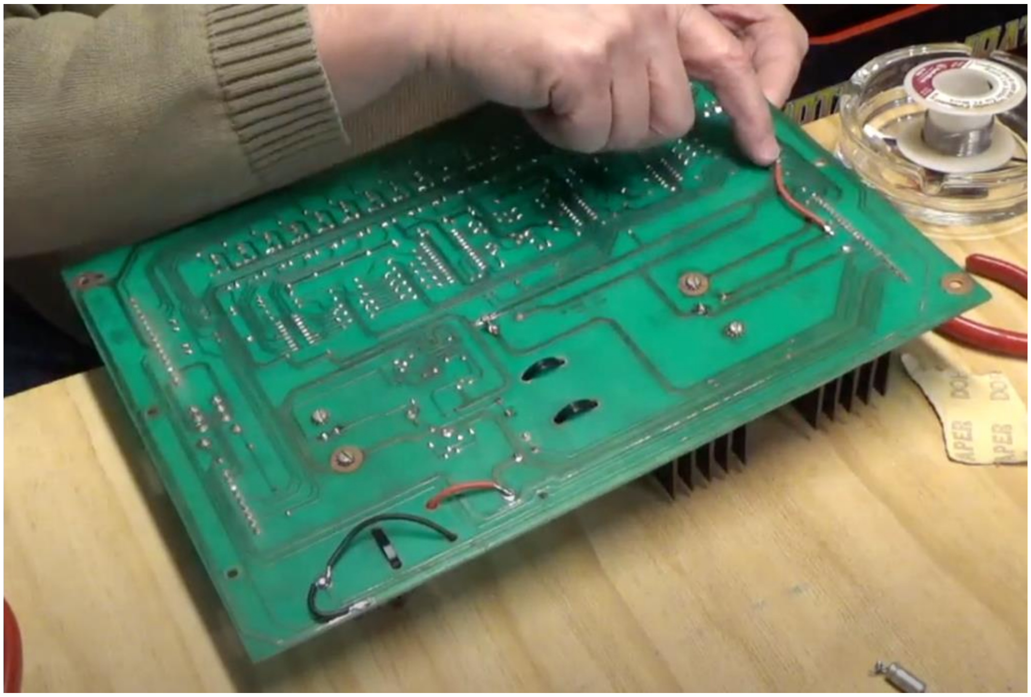


Figure 12

If you have an Alltek voltage regulator board, you do not need to make any modifications. Just remove the looping wire, and you're ready to connect your speaker. But if you are still using the original board, you'll have to make this modification. In Figure 12 you can see I have soldered a wire on the back of the voltage regulator board, connecting TP1 to TP3. This is a recommended modification in any case. TP1 and TP3 are directly connected to J3 pins 13 and 25, and so the loop connecting these pins can be removed.

See this video for additional information: www.youtube.com/watch?v=VWA-4wWTFIQ

Make a Connector

Next, you need to make a special connector (see Figure 13), starting with what is known as a USB Female Pigtail connector. It has a regular USB female plug on one end, and bare wires on the other. The red wire is your 5-volt line. Crimp a standard 0.1 inch connector to the red wire. The black wire is your ground. There are no spare ground connectors on J3, but we can connect this wire to ground at the corner of the voltage regulator board. Solder a metal washer to the black wire.

There may be two additional wires, that are usually green and white. These are data lines, and not needed for this purpose. Clip them off or tie them back out of the way.

In Figure 14, you can see the connector installed. The red wire is connected into pin 25 of J3, and the soldered washer is held under the corner screw of the board. I have placed it on top of the board, but you may need to place it under the board to get a good connection. Notice also that I have secured the connector's wire using a cable clamp, to eliminate any strain on the connections.

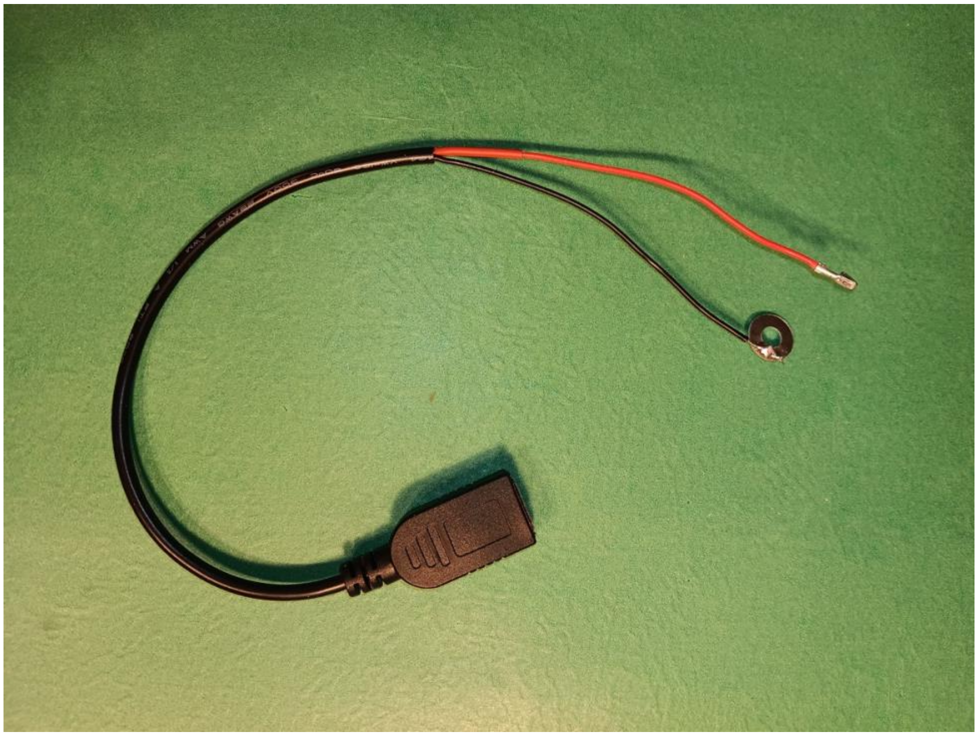


Figure 13

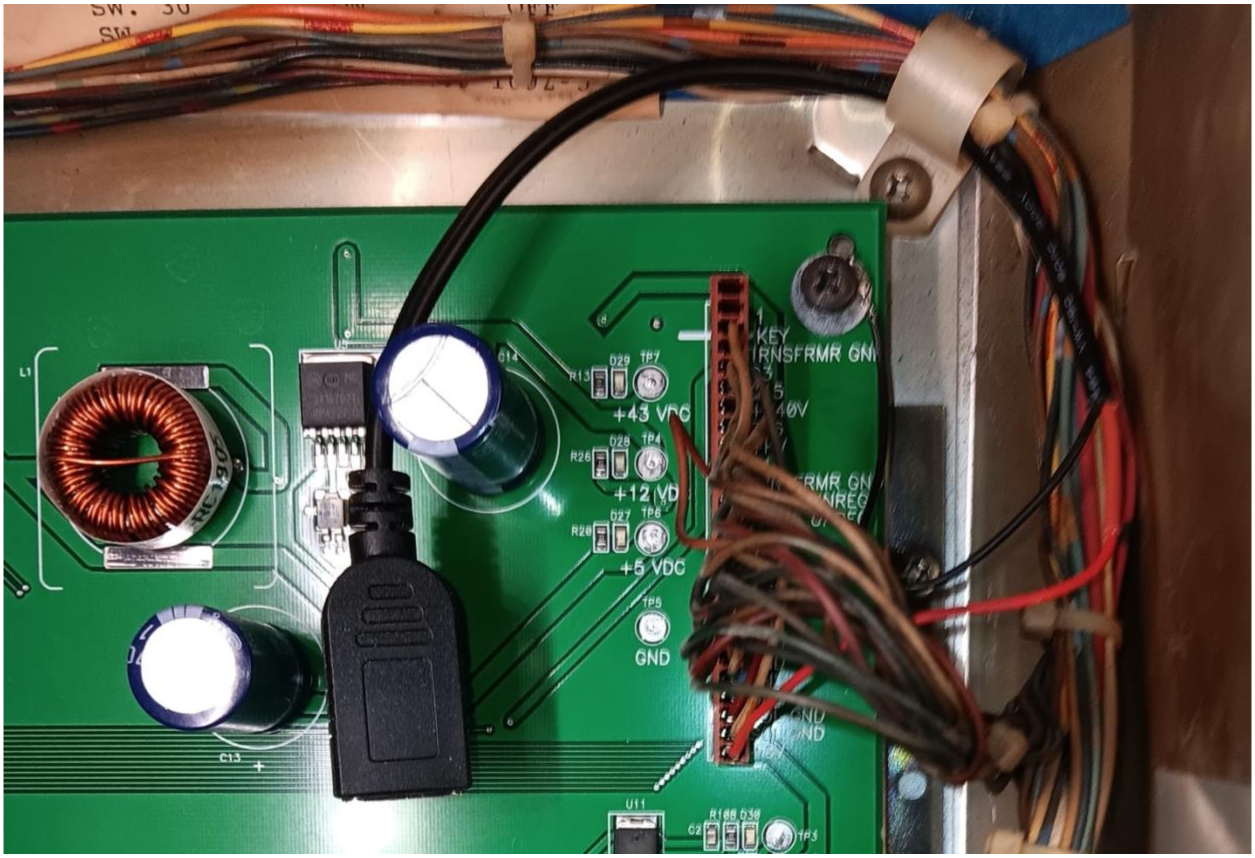


Figure 14

Figure 15

The Final Setup

Figure 15 shows the entire setup, including the Arduino board on the MPU, connected to the WAV Trigger board at the top of the backbox, with the speaker bar at the bottom plugged into the WAV Trigger with a 3.5mm stereo jack (bottom right of the board), and plugged into our pigtail connector with a USB connector for power (above the voltage regulator board).

Other wires you can see in the photo include:

1. A pair of green and white wires attached to the Arduino's "Switch" connector, curling up and over the top before exiting through the back vents.
2. A black cable connected to the Arduino's programming port, coiled around the top of the circuit board bracket before going up and over the top to the back vents.

You will likely not want the speaker to be inside the backbox! It is a simple task at this point, though, to move the speaker bar to the top of the backbox, and snake the wires through the vents at the back, into the backbox, to make the necessary connections.

Keen observers will note that the speaker wire from my original sound board is disconnected. This is necessary to prevent this board from playing unwanted sound during games. I have since added a switch on the back of the pinball that connects or disconnects the ground wire from this sound board to the original speaker. In fact, it is the same switch I use to turn the Arduino off and on (a 6-pin, 2-position toggle switch, connected to the Arduino's "Switch" connector). With the switch down, the Arduino is disconnected and the old speaker is connected. I can then play the original game with the original sound from the original speaker. Or with the switch up, The Arduino board controls the game with sound from the speaker bar, and the old speaker is disconnected (see Figure 16).

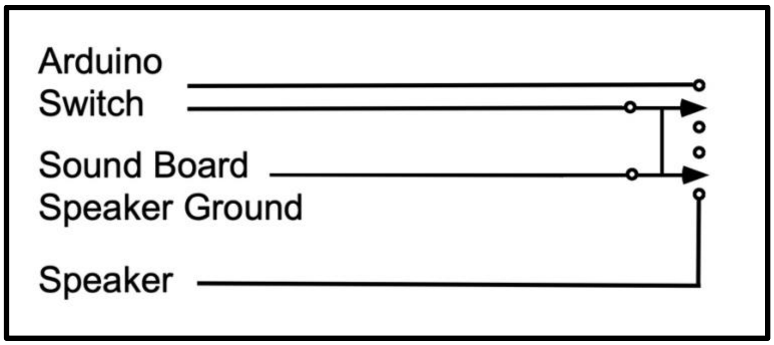


Figure 16

For additional information please see:

https://learn.sparkfun.com/tutorials/wav-trigger-hookup-guide-v11/all
https://cdn.sparkfun.com/assets/1/c/9/a/9/WT_UserGuide_20230114.pdf
https://www.youtube.com/watch?v=VWA-4wWTFIQ

Shopping List

This section pulls together everything mentioned in the manual that you might have to purchase in order to complete the project. Note that, you might get away with just purchasing and installing the Arduino plus WAV Trigger. But in case you really get deep into the project, these are all the items you might need!

Basic Requirements

Purchase Arduino with WAV Trigger: pinside.com/pinball/shops/shop/1304-roygbev-pinball/13777-star-trek-arduino-upgrade-with-new-rules-and-sounds
Or Build Your Own: www.pinballrefresh.com/blog/how-to-install-on-your-machine
Purchase Arduino IDE: www.arduino.cc/en/software
Purchase WAV Trigger: www.robertsonics.com/wav-trigger (if not included above)

Watch the following YouTube video for step-by-step instructions on purchasing and adding Star Trek music to your pinball: https://youtu.be/CJBlf6C11SE?si=8CX0rF0vCIO9eKpC

Programming Cable: You need a data cable (not a power cable!) with a USB micro connector on one end, and a USB A or C on the other end, whichever your computer accepts.

Micro-SD Card: The WAV Trigger requires a 32GB or less micro-SD card.

WAV Trigger Requirements

6-pin IDC ribbon cable: Single-Row / 6 position / Female to Female / ten inches or longer.
0.1" 6-pin vertical male through-hole header: Often sold with 40 pins, which can then be trimmed.
Computer speaker bar: Must have 3.5mm stereo jack and USB power connector.
Four narrow screws, and eight plastic washers.

Speaker Power Requirements

Old USB phone charger or
USB female pigtail connector
Standard 0.1" crimp connector
Metal washer

More Stuff

Micro-SD to USB card reader
Double pole, double throw (6-pin, 2-position) toggle switch
Wire
Soldering iron
Solder